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Ciao, I'm Fabio, I Am Very Interested in what Makes Topology Work

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Submitted in part fulfilment of the requirements for the degree of
Doctor of Philosophy in Physics of Imperial College of Science,
Technology and Medicine, December 2023

Abstract

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Chapter 1

Introduction

1.1 Go on a different lattice

1.2 Quantum Hall and Topological Insulators

1.2.1 Initial motivation - The quantum hall effect

1.2.2 The TKNN Formula and connection to Chern number

1.3 Spin Liquids

1.3.1 background - anderson RVB, frustrated magnets, X-G Wen and co

1.3.2 Kitaev Model

1.3.3 Physical materials

1.4 Thesis Outline

Chapter 2

Lattices and Their Transformations

2.1 Voronoi Lattices

2.2 LLoyd's Algorithm

2.3 Dual Lattices

2.4 Vertex Expansion

Chapter 3

Chern Number and Topological Insulators

Paragraph introducing the topics covered in this section. Points to hit:

- Periodic translational hamiltonians can be classified according to the chern number of their bands, where nonzer chern number implies the existence of gapless edge modes in open boundaries.
- This classification has been immensely successful for understanding a lot of materials eg: QHE, Spin Hall effect etc
- The classification relies on a notion of momentum space - in order to construct a manifold of states on which to define parallel transport.
- Thus it doesn't work for systems without k space - eg. open boundaries/disorder
- Local markers are an attempt at constructing a real space version of this
- We will explain all the prerequisites for constructing a local marker, as well as discussing their limitations.

Throughout this chapter and the next we will focus on non-interacting electrons on a lattice, thus let us begin by defining our Hilbert space in a first-quantised framework. We will work with the most general definition of a non-interacting tight-binding Hamiltonian in two dimensions. The Hilbert space $\mathcal{H}$ contains a set of N sites, where site j is at a position $\mathbf{r}_j$, and has η internal degrees of freedom. For example, our sites could correspond to atoms, and the states at each site could represent individual orbitals around an atom. We will not assume that our sites are arranged in a regular lattice – we will also want to model amorphous structures – thus, a state in this system can be written as

$$|\psi\rangle = \sum_j c_{\mathbf{r}_j, \alpha} |\mathbf{r}_j\rangle \otimes |\alpha\rangle, \quad (3.0.1)$$

where $c_{\mathbf{r}_j, \alpha}$ are a set of complex coefficients and $\alpha \in \{1 \dots \eta\}$ labels the internal degree of freedom. A general Hamiltonian in this context takes the form

$$H = \sum_{j, k, \alpha, \beta} H_{j, k, \alpha, \beta} |\mathbf{r}_j\rangle \langle \mathbf{r}_k| \otimes |\alpha\rangle \langle \beta|. \quad (3.0.2)$$

If the system has translational symmetry $\{\mathbf{r}_j\}$ will define a regular lattice with periodic boundaries and the Hamiltonian is invariant under translations – we can of course diagonalise the Hamiltonian in a basis of k -states,

$$|\mathbf{k}\rangle = \frac{1}{\sqrt{N}} \sum_{\mathbf{r}_j} e^{-i\mathbf{k}\mathbf{r}_j} |\mathbf{r}_j\rangle. \quad (3.0.3)$$

For a finite system the k -states are quantised to $k_a = 2\pi n/L_a$, where L_a is the length of the system in direction $a \in \{x, y\}$, whereas in the case of an infinite system k can take any value in the interval $[0, 2\pi]$.

3.1 Abelian Berry Phase and the Chern Number

In this section we introduce the mathematical construction of Berry phase, Berry flux and Chern number. We will present these ideas in two different contexts. In §3.1.1 we introduce the simplest possible story that captures the essentials of Berry phase and Chern number: that of a discretised lattice of quantum states. Here we will miss many of the details that make such concepts relevant to physical systems, however the picture is sufficiently simple as to provide a practical intuition for how to calculate these quantities. In §3.1.2, we will do things *properly*, by considering the effect on the band structure of some Hamiltonian that is changed adiabatically over some smooth parameter space. We will derive the exact form of the Chern number in this context and relate it to the TKNN invariant [1]. Finally, in §3.1.3 we give a brief example of such a calculation, demonstrating how these concepts apply to a two-state quantum system.

3.1.1 The Simplest Story – A Discrete Lattice of States

In quantum mechanics a state is defined as an equivalence class of vectors, where two states that differ only by a complex phase are considered physically equivalent,

$$|\psi\rangle \sim e^{-i\alpha} |\psi\rangle. \quad (3.1.1)$$

Thus, we seem to have a degeneracy built into our description of physics. This means that one can define a gauge transformation – multiplying a state with a complex phase – that is guaranteed not to affect the system in any ‘real’ measurable way. Any derived quantity must be invariant under gauge transformations in order to be considered physical. Thus, one would expect any quantity that depends on the state’s phase to be unphysical.

Similarly, considering some arbitrary set of states in the same Hilbert space, $\{|\psi_j\rangle\}$, we may define a local gauge transformation where we apply a different phase to each individual state,

$$|\psi_j\rangle \rightarrow e^{-i\alpha_j} |\psi_j\rangle. \quad (3.1.2)$$

Again, the physical meaning of each state is unaffected, so we expect that any physical quantity calculated from these states must be invariant under the transformation. Thus one might expect that any *phase-like* quantity calculated from these states should have no physical relevance. For

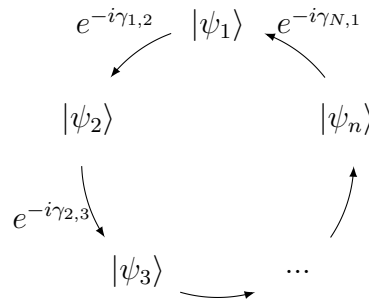
example, let us consider the relative phase between state j and k , $\gamma_{j,k}$, defined according to

$$e^{-i\gamma_{j,k}} = \frac{\langle \psi_j | \psi_k \rangle}{|\langle \psi_j | \psi_k \rangle|}. \quad (3.1.3)$$

Clearly this quantity is invariant if all states pick up the same phase, however it is not invariant under a local transformation, changing as

$$e^{-i\gamma_{j,k}} \rightarrow e^{-i(\gamma_{j,k} + \alpha_j - \alpha_k)}, \quad (3.1.4)$$

and so cannot be a physical observable. In order to construct a gauge-invariant object, let's pick a set of n (≥ 3) states $|\psi_j\rangle$ and arrange them in a loop.



We can now define a new quantity called the Berry phase:

$$e^{-i\gamma_B} = \frac{\langle \psi_1 | \psi_2 \rangle \langle \psi_2 | \psi_3 \rangle \dots \langle \psi_N | \psi_1 \rangle}{|\langle \psi_1 | \psi_2 \rangle \langle \psi_2 | \psi_3 \rangle \dots \langle \psi_N | \psi_1 \rangle|} \quad (3.1.5)$$

Another way of writing this would be

$$\gamma_B = -\arg \text{Tr} [\hat{\psi}_1 \hat{\psi}_2 \dots \hat{\psi}_N], \quad (3.1.6)$$

where the trace is over the whole system, and $\hat{\psi}$ is shorthand for the density operator

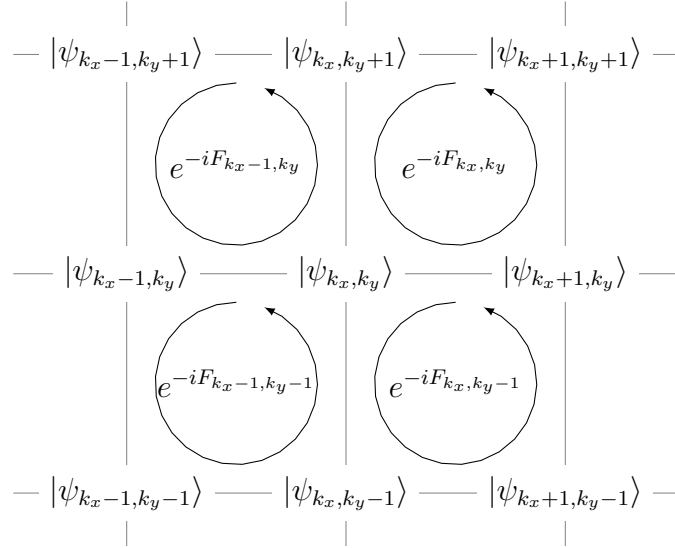
$$\hat{\psi}_i = |\psi_i\rangle \langle \psi_i|. \quad (3.1.7)$$

This quantity is clearly gauge-invariant, since any phase applied to a state $|\psi_m\rangle$ will show up twice, once from $\langle \psi_m | \psi_{m+1} \rangle$ and once from $\langle \psi_{m-1} | \psi_m \rangle$. The phase will have opposite sign in each of these expressions so they cancel out. Furthermore, we can see that the Berry phase around a loop is simply the sum of all the relative phases around that loop modulo 2π .

$$\gamma_B = \sum_{n=1}^N \gamma_{n,n+1} \mod 2\pi \quad (3.1.8)$$

Berry Flux

Now suppose that we have a large number of states arranged in a 2D grid. These states $|\psi_{\mathbf{k}}\rangle$ are labelled according with a pair of indices k_x and k_y . In general, this parameter $\mathbf{k}$ will refer to the crystal momentum, however the construction is equally valid for any other parameter that defines a 2D grid of quantum states.



We introduce a new quantity called the Berry flux. This is defined as the Berry phase around a plaquette on our grid,

$$F_{\mathbf{k}} = -\arg \text{Tr} \left[\hat{\psi}_{k_x, k_y} \hat{\psi}_{k_x+1, k_y} \hat{\psi}_{k_x+1, k_y+1} \hat{\psi}_{k_x, k_y+1} \right]. \quad (3.1.9)$$

$F_{\mathbf{k}}$ is just the sum (mod 2π) of the relative phases along each edge of the plaquette starting at point $\mathbf{k}$. Thus, the Berry phase around some arbitrary loop containing multiple plaquettes can be found by summing the value of $F_{\mathbf{k}}$ for every plaquette contained in that region. Each relative phase flips sign depending on which way round it is evaluated. Any internal edge is shared between two plaquettes and so will appear twice in the sum of Berry fluxes, with opposite signs. This means all internal edges have their contribution to the total phase cancelled out and we are left with the sum of the relative phases only around the boundary of our region.

$$\exp \left[-i \sum_{\text{inside region}} F \right] = \exp \left[-i \sum_{\text{boundary}} \gamma \right] \quad (3.1.10)$$

This resembles Stokes' law, however there is a subtlety worth mentioning. All we have required is that the complex exponential of each sum is the same. This does **not** mean that the sum of the Berry flux inside a region is equal to the sum of relative phases around the boundary, since they are allowed to differ by an arbitrary multiple of 2π .

add a touch more here - it was obvious to k but maybe not to everyone

Chern Number

Now let's assume that our grid of $\mathbf{k}$ points has periodic boundaries, with $1 \leq k_x \leq N_x$ and $1 \leq k_y \leq N_y$. That is, the lattice is a discretisation of the torus – a closed manifold. The product of all the Berry flux phases is now always equal to 1,

$$\prod_{k_x=1}^{N_x} \prod_{k_y=1}^{N_y} e^{-iF_{k_x, k_y}} = 1. \quad (3.1.11)$$

Here because every edge in our grid is now an internal edge and thus they cancel out. Again, as before, this does not mean that the sum of all Berry fluxes in our system vanishes. It means that the sum must always equal some multiple of 2π . This sum gives us the Chern number, $\mathcal{Q}$,

$$\mathcal{Q} = \frac{1}{2\pi} \sum_{k_x, k_y} F_{k_x, k_y}. \quad (3.1.12)$$

$\mathcal{Q}$ is necessarily restricted to integer values, since a non-integer value would contradict equation 3.1.11.

3.1.2 Topology of a General Hamiltonian

Now that we have a basic intuition for what the Chern number is, let's see how this applies to a real physical system. We start by defining some arbitrary Hamiltonian $H(\mathbf{R})$, which depends on some smooth parameter $\mathbf{R} \in \mathcal{R}$.

For every value of $\mathbf{R}$, the Hamiltonian will generally have some set of eigenstates $|\psi_n(\mathbf{R})\rangle$ with corresponding energies ε_n . As before, these eigenstates are only strictly defined up to a phase. Thus let consider a single state $|\psi(\mathbf{R})\rangle$ and define a local gauge transformation as applying some $\mathbf{R}$ -dependent phase $\alpha(\mathbf{R})$ to all of the states over this band.

$$|\psi(\mathbf{R})\rangle \rightarrow e^{-i\alpha(\mathbf{R})} |\psi(\mathbf{R})\rangle. \quad (3.1.13)$$

In order for this to be well-behaved we will need to use a *smooth gauge*. This is a choice of gauge that ensures that the $\mathbf{R}$ -derivative of a state, $\nabla_{\mathbf{R}} |\psi(\mathbf{R})\rangle$ is always well-defined – that is, $\alpha(\mathbf{R})$ does not jump around discontinuously. In general it is not always possible to pick a gauge that is globally smooth over all of $\mathcal{R}$, however it *is* always possible to pick a locally smooth gauge around the neighbourhood of some point in $\mathcal{R}$.

Let us define relative phase between a state at $\mathbf{R}$ and another state that is infinitesimally displaced from it, at $\mathbf{R} + d\mathbf{R}$,

$$e^{-i\Delta\gamma} = \frac{\langle \psi(\mathbf{R}) | \psi(\mathbf{R} + d\mathbf{R}) \rangle}{|\langle \psi(\mathbf{R}) | \psi(\mathbf{R} + d\mathbf{R}) \rangle|}. \quad (3.1.14)$$

In the limit of small $d\mathbf{R}$ we can express $|\psi(\mathbf{R} + d\mathbf{R})\rangle$ to first order in $d\mathbf{R}$ as

$$|\psi(\mathbf{R} + d\mathbf{R})\rangle = |\psi(\mathbf{R})\rangle + d\mathbf{R} \cdot \nabla_{\mathbf{R}} |\psi(\mathbf{R})\rangle. \quad (3.1.15)$$

Which means that

$$\langle \psi(\mathbf{R}) | \psi(\mathbf{R} + d\mathbf{R}) \rangle = 1 + \langle \psi(\mathbf{R}) | \nabla_{\mathbf{R}} |\psi(\mathbf{R})\rangle \cdot d\mathbf{R}. \quad (3.1.16)$$

Now, noticing that $\langle \psi(\mathbf{R}) | \nabla_{\mathbf{R}} |\psi(\mathbf{R})\rangle$ is pure imaginary, we can show that

$$\Delta\gamma = \mathbf{A} \cdot d\mathbf{R}, \quad (3.1.17)$$

$$\mathbf{A}(\mathbf{R}) = i \langle \psi(\mathbf{R}) | \nabla_{\mathbf{R}} |\psi(\mathbf{R})\rangle. \quad (3.1.18)$$

The quantity multiplying $d\mathbf{R}$ is the Berry connection $\mathbf{A}(\mathbf{R})$. Like the relative phase in the previous section, it is not a gauge-invariant quantity. Under a gauge transformation it changes

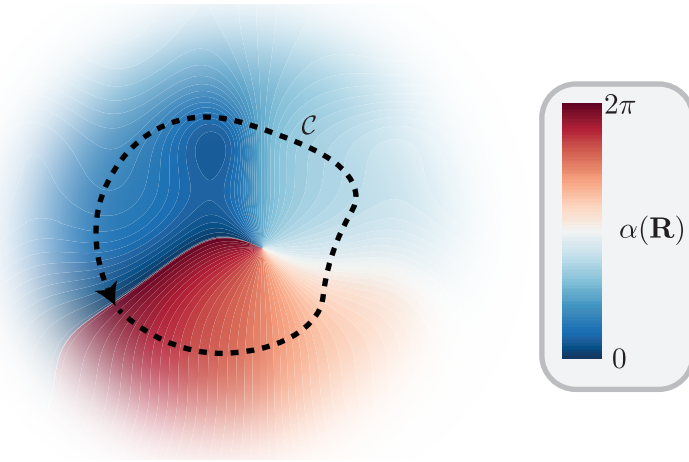


Figure 3.1: An example of a non-smooth gauge transformation designed to change the value of the Berry phase around the path $\mathcal{C}$ – marked with a dashed line – by an addition of 2π . Only a small region of $\mathbf{R}$ -space is shown, with the local phase applied shown by the colormap.

according to

$$\mathbf{A}(\mathbf{R}) \rightarrow \mathbf{A}(\mathbf{R}) - \nabla_{\mathbf{R}}\alpha(\mathbf{R}) \quad (3.1.19)$$

Now let us suppose that we slowly transform our Hamiltonian around some closed path $\mathcal{C}$ in $\mathbf{R}$ -space and keep track of the phase picked up by a given state.¹ In analogy with eqn. 3.1.8, the Berry connection allows us to calculate the total phase accumulated over this route, according to

$$\gamma(\mathcal{C}) = \oint_{\mathcal{C}} \mathbf{A} \cdot d\mathbf{R}. \quad (3.1.20)$$

The exponential of the phase $e^{-i\gamma(\mathcal{C})}$ around the loop *is* gauge invariant, one can see this intuitively by realising that it is identical to the Berry phase from the discrete case, but in the limit where the discrete steps become arbitrarily close together. However, the Berry phase $\gamma(\mathcal{C})$ itself is not strictly gauge invariant, since one could imagine introducing a gauge transformation that introduces a branch cut around the path $\mathcal{C}$. After traversing the whole path, the state at the end of the loop would have picked up a total phase 2π , meaning that under a gauge transformation we can change the berry phase according to

$$\gamma(\mathcal{C}) \rightarrow \gamma(\mathcal{C}) + 2\pi n \text{ for } n \in \mathbb{Z}. \quad (3.1.21)$$

Berry Curvature

Let us now define a quantity equivalent to the Berry flux from the previous discussion. Starting with eqn. 3.1.20, we use Stokes' theorem to transform it into a surface integral over a surface $\mathcal{S}$

¹Many sources (e.g. [2]) present this picture as an *adiabatic* transformation, where one has to do a little extra work to keep track of the energy-dependent phase that a state picks up $\sim e^{-i\varepsilon t}$ alongside the topological phase determined by $\mathbf{A}$. We won't bother, and instead just consider comparing the *instantaneous eigenstates* at different values of $\mathbf{R}$.

bounded by the curve $\mathcal{C}$. We will consider the case for a two and three dimensional parameter space $\mathcal{R}$, but of course this argument generalises to arbitrary dimensions.

$$\text{In 2D: } \gamma(\mathcal{C}) = \int_S d^2 R (\partial_x A_y(\mathbf{R}) - \partial_y A_x(\mathbf{R})), \quad (3.1.22)$$

$$\text{In 3D: } \gamma(\mathcal{C}) = \int_S d\mathbf{S} \cdot (\nabla_{\mathbf{R}} \times \mathbf{A}(\mathbf{R})), \quad (3.1.23)$$

where we use the shorthand $\partial_\alpha = \frac{\partial}{\partial R_\alpha}$. Thus we may define the Berry curvatures,

$$\Omega(\mathbf{R}) = \partial_x A_y(\mathbf{R}) - \partial_y A_x(\mathbf{R}) \quad (3.1.24)$$

$$\mathbf{\Omega}(\mathbf{R}) = \nabla_{\mathbf{R}} \times \mathbf{A}(\mathbf{R}). \quad (3.1.25)$$

Two Useful Ways to Re-express the Berry Curvature — Now let us focus only on the 2D case and reformulate this expression in two different ways.

1. The first way finds the curvature in terms of the rate of change of the Hamiltonian only. Writing this quantity out explicitly in terms of the states $|\psi(\mathbf{R})\rangle$, we arrive at

$$\Omega(\mathbf{R}) = i \langle \partial_x \psi(\mathbf{R}) | \partial_y \psi(\mathbf{R}) \rangle + h.c. \quad (3.1.26)$$

We can insert a resolution of the identity $\mathbb{1} = \sum_n |\psi_n\rangle \langle \psi_n|$ to obtain

$$\Omega(\mathbf{R}) = i \sum_{\psi_n \neq \psi} \langle \partial_x \psi | \psi_n \rangle \langle \psi_n | \partial_y \psi \rangle + h.c. \quad (3.1.27)$$

Using the identity,

$$\langle \psi_n | \partial_\alpha \psi \rangle = \frac{\langle \psi_n | \partial_\alpha H | \psi \rangle}{\epsilon - \epsilon_n}, \text{ for } \psi \neq \psi_n, \quad (3.1.28)$$

we can re-express the Berry curvature in the form

$$\Omega(\mathbf{R}) = i \sum_{\psi_n \neq \psi} \frac{\langle \psi | \partial_x H | \psi_n \rangle \langle \psi_n | \partial_y H | \psi \rangle}{(\epsilon - \epsilon_n)^2} + h.c. \quad (3.1.29)$$

2. Note that the definition provided for the Berry curvature (eqn. 3.1.26) is only well-defined across $\mathcal{R}$ when the state remains fully gapped. That is, no other states ever have the same energy as $|\psi(\mathbf{R})\rangle$. When another state energy crosses that of $|\psi\rangle$, the individual states become ill-defined and so it becomes challenging to write down a natural derivative for the states. With this in mind, let us consider re-expressing the Berry curvature in terms of the single state projector $\hat{\psi} = |\psi\rangle \langle \psi|$,

$$i \langle \partial_x \psi | \partial_y \psi \rangle + h.c. = i \text{Tr} \left[\hat{\psi} \partial_x \hat{\psi} \partial_y \hat{\psi} \right] + h.c. \quad (3.1.30)$$

For a single state, this is identical to the regular Berry curvature, however the utility of this expression becomes evident when we construct it for a projector onto a set of states $P = \sum_{i \in \text{band}} |\psi_i\rangle \langle \psi_i|$. Since ∂P is still well-defined when the states in P are

degenerate, we can write an expression for the combined berry curvature of a set of states that can handle degeneracy (provided no states outside P become degenerate with P).

$$\Omega_P = i\text{Tr} [P\partial_x P\partial_y P] + h.c. \quad (3.1.31)$$

This calculation has one important caveat. It is only possible to apply Stokes' theorem when $\mathbf{A}$ is smooth over the region $\mathcal{R}$. This means that we need to have already picked a smooth gauge in the region we are integrating over to have a guarantee that eqns. 3.1.22 and 3.1.22 hold. If we transform to a non-smooth gauge then it is possible that the Berry phase will change by a multiple of 2π , however the Berry curvature is truly gauge invariant. This can be seen from eqn. 3.1.29, where the Berry curvature is defined without the differentiation of states, so is invariant under even a non-smooth gauge transformation. Thus in general we only have the guarantee that

$$e^{-i\gamma(\mathcal{C})} = e^{\int_S d^2 R \Omega(\mathbf{R})}. \quad (3.1.32)$$

Chern Number

As in §3.1.1, we make the assertion that the $\mathbf{R}$ -space lives on the torus, a closed manifold. The Chern number can then be defined as the integral of the Berry curvature across the whole manifold,

$$\mathcal{Q} = \frac{1}{2\pi} \int d^2 R \Omega. \quad (3.1.33)$$

As in the discrete case, the Chern number here is restricted to only take integer values. The argument follows the same lines as previously, where we can use our modified form of Stokes theorem (eqn. 3.1.32) to show that the Chern number is equivalent to the Berry phase around a loop enclosing no area, implying that

$$e^{\int d^2 R \Omega(\mathbf{R})} = 1, \implies \mathcal{Q} \in \mathbb{Z} \quad (3.1.34)$$

3.1.3 An Example – The Two-Level System

Let us illustrate these concepts with the example of a two-state system. Our first step is to parametrise the possible Hamiltonians in such a system using a vector $\mathbf{d}$. We will then solve this Hamiltonian for the positive and negative energy eigenstates. This allows us to look at how the eigenstates change as we smoothly deform the Hamiltonian, exploring the parameter space of $\mathbf{d}$. By examining how the eigenstates of $\hat{H}$ depend on the parametrisation, we can calculate the Berry curvature of the positive and negative energy states. We then connect this with our understanding of the Bloch states in a periodic quantum system. We discuss how the bulk Hamiltonian of a periodic system corresponds to the embedding of a closed surface in $\mathbf{d}$ -space and describe how to calculate the Chern number of the bands in such a system.

The Hilbert space of a two-level system is $\mathbb{C}^2$. This means that any Hamiltonian we construct must be a 2×2 Hermitian matrix. We will parametrise this in terms of a Pauli vector, $\mathbf{d}$,

$$H = \mathbf{d} \cdot \hat{\sigma} = d_i \sigma^i. \quad (3.1.35)$$

where the σ^i are the Pauli matrices, which form a complete basis for 2×2 traceless hermitian matrices. Of course, we could include a multiple of the identity to span the full space of Hermitian matrices, however this would have no effect on the topology so we shall omit it.

The set of traceless Hamiltonians is thus a manifold in itself, $\{H\} \cong \mathbb{R}^3$, since $\mathbf{d} \in \mathbb{R}^3$. The first thing we will do is to express the vector $\mathbf{d}$ in standard spherical polar coordinates, giving us a Hamiltonian that looks like

$$H = |\mathbf{d}| \begin{pmatrix} \cos(\theta) & e^{-i\phi} \sin \theta \\ e^{i\phi} \sin \theta & -\cos(\theta) \end{pmatrix} \quad (3.1.36)$$

This matrix has $H^2 = |\mathbf{d}|^2 \mathbb{1}$, so its eigenvalues are $\pm |\mathbf{d}|$. One can show that its eigenvectors are

$$|+\mathbf{d}\rangle = \begin{pmatrix} e^{-i\phi/2} \cos(\theta/2) \\ e^{i\phi/2} \sin(\theta/2) \end{pmatrix}, \quad |-\mathbf{d}\rangle = \begin{pmatrix} -e^{-i\phi/2} \sin(\theta/2) \\ e^{i\phi/2} \cos(\theta/2) \end{pmatrix}. \quad (3.1.37)$$

We have thus arrived at a set of states, $|\pm\mathbf{d}\rangle$ that depend on some parameter $\mathbf{d}$. This means that we can apply the techniques of the last section. From eqns. 3.1.18 and 3.1.23 we know that the Berry connection and Berry curvature of the $|\pm\rangle$ states are given by

$$\mathbf{A}^\pm(\mathbf{d}) = i \langle \pm\mathbf{d} | \nabla_{\mathbf{d}} | \pm\mathbf{d} \rangle \quad (3.1.38)$$

$$\Omega^\pm(\mathbf{d}) = \nabla_{\mathbf{d}} \wedge \mathbf{A}^\pm(\mathbf{d}). \quad (3.1.39)$$

Using this equation it is possible to directly calculate the Berry curvature, arriving at the following expression:

$$\Omega^\pm = \pm \frac{\hat{n}_{\mathbf{d}}}{2d^2} \quad (3.1.40)$$

where $\hat{n}_{\mathbf{d}}$ is a unit vector pointing in the direction of $\mathbf{d}$.

The Berry curvature is equivalent to a monopole field radiating out from the origin in $\mathbf{d}$ -space, in analogy with electrostatics. Thus, the Berry phase of the $|+\mathbf{d}\rangle$ state around some path in $\mathbf{d}$ -space is given by

$$\gamma^+(C) = \oint_C \mathbf{A}^+(\mathbf{d}) \cdot d\mathbf{d}. \quad (3.1.41)$$

Re-expressing this in terms of the Berry curvature gives us

$$\gamma^+(C) = \int_{\mathcal{R}} \Omega^+(\mathbf{d}) \cdot d\mathbf{S}, \quad (3.1.42)$$

where the integral is over a region $\mathcal{R}$ that has the curve $\mathcal{C}$ as its boundary. Since the Berry curvature is just a monopole field with charge $\frac{1}{2}$, this integral ends up evaluating half the solid angle subtended by the curve $\mathcal{C}$.

Chern Number

Now we can connect this with our understanding of Bloch states. For a two-dimensional translationally symmetric system, the Hamiltonian can be diagonalised according to

$$H = \sum_{\mathbf{k}} |\mathbf{k}\rangle \langle \mathbf{k}| \otimes H(\mathbf{k}). \quad (3.1.43)$$

If the unit cell contains two sites, then $H(\mathbf{k})$ is a 2×2 matrix that depends on $\mathbf{k}$. We can write it in terms of a $\mathbf{d}$ vector that depends on $\mathbf{k}$.

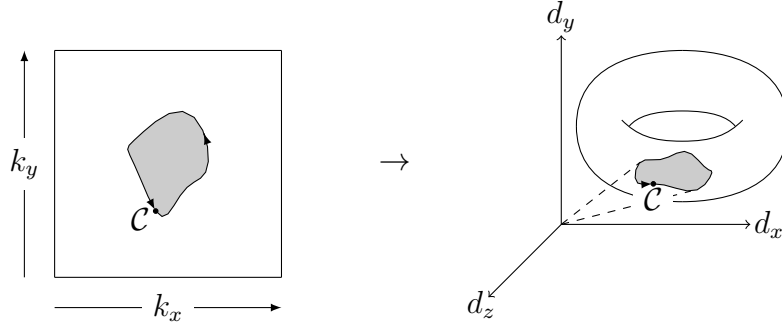
$$H(\mathbf{k}) = \mathbf{d}(\mathbf{k}) \cdot \boldsymbol{\sigma}. \quad (3.1.44)$$

$\mathbf{k}$ lives on the surface of a torus, since k_x and k_y are periodic. Furthermore, $\mathbf{d}$ (and by extension H) live in the space $\mathbb{R}^3$. Thus, the mapping $\mathbf{d}$ is

$$\mathbf{d} : S^1 \times S^1 \rightarrow \mathbb{R}^3. \quad (3.1.45)$$

The map is embedding a two-dimensional surface (the torus) into three-dimensional flat space ($\mathbb{R}^3$). There are no limits on how you can do this embedding, the torus can take any shape provided the embedding respects the topology of the Brillouin zone.

To calculate the Berry phase around a loop in $\mathbf{k}$ -space, all we have to do is look at the corresponding loop in $\mathbf{d}$ -space. Then we integrate the Berry connection over some surface that has this loop as its boundary. The Berry connection actually looks like a radiating monopole so we can simplify further, knowing that we just need to find the solid angle subtended by the curve around the origin and divide it by two.



What about the Chern number? To calculate this all we need to do is integrate the Berry curvature across the whole surface of the torus in $\mathbf{d}$ -space. If the embedding of the torus does not contain the origin, then the value of this integral is necessarily zero.

Now suppose the torus is embedded such that it contains the origin, then we will have a surface that catches all of the Berry flux on its inside and the solid angle subtended is 4π ! This means that the Chern number will be $+1$. Furthermore, you could imagine wrapping the torus around the origin twice, so that it effectively contains two origins, now the Chern number will be $+2$. You could turn it inside out and then wrap it around the origin too, that would give $\mathcal{Q} = -1$ and so on...

3.1.4 Final Notes

this section needs a bit more formal equation-y stuff

We close this section with an explanation of the consequences of Chern number in topological insulators.

Smoothly Deforming the Hamiltonian

Now suppose that we have a pair of quantum systems that live in the same Hilbert space, described by $\hat{H}_1$ and $\hat{H}_2$. Suppose both of these Hamiltonians represent insulating materials. This means that the bands in these materials are separated across all of the Brillouin zone by a band gap. We might want to ask the question: *Is it possible to smoothly deform $\hat{H}_1$ into $\hat{H}_2$ without ever crossing a point where the bands touch and the material becomes conductive?*

The answer to this is that it is only possible if the bands of $\hat{H}_1$ and $\hat{H}_2$ have the same Chern numbers. This is because the Chern number is always restricted to have integer value, it cannot continuously change as we smoothly deform the Hamiltonian. In order to change the Chern number of a band we must somewhere cross a point where it is impossible to define a Chern number at all, allowing it to jump discontinuously when we leave that point. This is precisely what happens when two bands touch, the description of a single band becomes ill-defined and our calculation for Chern number stops working.

We can illustrate this with the example in §3.1.3. Smoothly deforming our Hamiltonian amounts to changing the shape of the torus embedded in $\mathbf{d}$ -space. If any point on the surface of the torus touches the origin, we have a pair of zero-energy states appearing in our system, this means the bands touch and the system is conductive. If the band has a Chern number of one, this means that the torus contains the origin. If we wanted to smoothly deform this to have zero Chern number, we would need to move the torus such that the origin was now outside. At some point in this deformation the torus surface will have to touch the origin, creating a conducting Hamiltonian at that point.

Thus we can classify all Hamiltonians according to the Chern number of their bands. It is always impossible to smoothly deform a Hamiltonian into one with a different Chern number while remaining insulating. Similarly, it is always possible to smoothly deform two Hamiltonians into one another without closing a gap, provided that their bands have identical Chern numbers [3].

This allows us to make a rough justification for why conducting edge states must appear wherever we have a material with nonzero Chern number. Suppose we have a large material where the Hamiltonian is allowed to vary slowly over space, defined by $\hat{H}(\mathbf{r})$. Now let's say that in some region this Hamiltonian looks locally like it has non-zero Chern number, $\hat{H}_{\text{topological}}$, but in a different, distant region it looks locally like it has zero Chern number, $\hat{H}_{\text{trivial}}$. We expect that between these regions the Hamiltonian must be gradually transforming between $\hat{H}_{\text{topological}}$ and $\hat{H}_{\text{trivial}}$. As we have described, it is impossible for our Hamiltonian to make this transformation without creating a conducting surface somewhere between these two regions – edge states must appear! Loosely speaking, we can view the vacuum as a zero Chern number material, which gives us an intuitive picture for why we might expect materials with non-zero Chern number to always have edge states appearing on their surface, where the Hamiltonian has to transition to the vacuum.

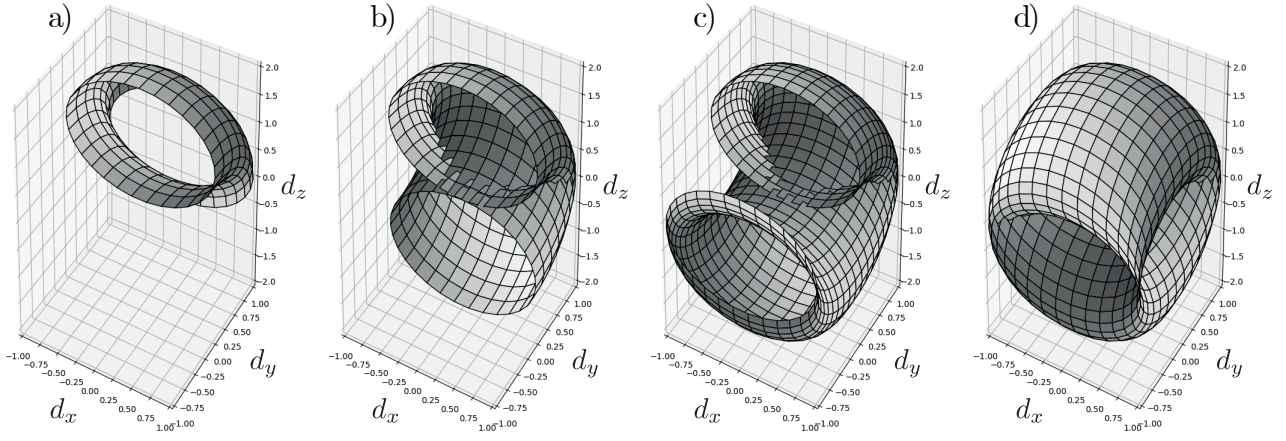


Figure 3.2: The embedding of the Brillouin zone in $\mathbf{d}$ -space for the QWZ model. This is shown in four progressive cutaways to ensure the internal structure can be seen. In all four images k_x spans the interval $[0, 2\pi)$, however in (a) k_y spans $[0, \frac{\pi}{2})$, in (b) k_y spans $[0, \pi)$, in (c) k_y spans $[0, \frac{3\pi}{2})$, and in (d) k_y spans $[0, 2\pi)$.

Chern Number and Globally Smooth Gauges

If you thought carefully about the gauge that we chose in equation 3.1.37 you might notice that it has a problem, There is actually a discontinuity at $\phi = \pi$ where $e^{i\phi}$ suddenly jumps as you cross over to $\phi = -\pi$. This is no oversight, it turns out that it is actually completely impossible to define a globally smooth set of states $|\psi(\mathbf{d})\rangle$ for any band that has nonzero Chern number!

This is because if $|\psi(\mathbf{d})\rangle$ were continuous everywhere, then the Berry connection would also have to be continuous everywhere. This would mean that we could invoke the exact Stokes' theorem when calculating the Chern number. Since the Berry curvature is integrated is over the whole of $\mathbf{k}$ -space it must have no boundary, $\mathcal{Q}$ must be equal to zero.

In this sense, the Chern number can be regarded as something like a measure of the amount of discontinuity that *must* be present regardless of the gauge you choose for the space.

3.2 The Qi-Wu-Zhang Model

Let us introduce the Qi-Wu-Zhang (QWZ) model as a paradigmatic example of a two-dimensional topological insulator with nearest-neighbour interactions [4, 5].

add some history on the model here

We work on a square lattice of unit cells indexed by a two dimensional lattice vector $\mathbf{m}$, where each cell contains two sites. The QWZ model is defined by three terms, a hopping between unit cells in the x -direction, a hopping in the y -direction, and a staggered on-site potential whose

strength is determined by a parameter u .

$$\begin{aligned}\hat{H} = & \sum_{\mathbf{m}} |\mathbf{m} + \mathbf{1}_x\rangle \langle \mathbf{m}| \otimes \frac{\sigma_z + i\sigma_x}{2} + h.c. \\ & + \sum_{\mathbf{m}} |\mathbf{m} + \mathbf{1}_y\rangle \langle \mathbf{m}| \otimes \frac{\sigma_z + i\sigma_y}{2} + h.c. \\ & + \sum_{\mathbf{m}} |\mathbf{m}\rangle \langle \mathbf{m}| \otimes u \sigma_z,\end{aligned}\tag{3.2.1}$$

where $\mathbf{1}_x$ and $\mathbf{1}_y$ represent the unit displacement in their respective directions,

$$\mathbf{1}_x = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad \mathbf{1}_y = \begin{pmatrix} 0 \\ 1 \end{pmatrix}.\tag{3.2.2}$$

Using Bloch's theorem, we can find the bulk Hamiltonian,

$$\hat{H}(\mathbf{k}) = \begin{pmatrix} \sin k_x \\ \sin k_y \\ u + \cos k_x + \cos k_y \end{pmatrix} \cdot \boldsymbol{\sigma}.\tag{3.2.3}$$

Here, k_x and k_y determine our position in the Brillouin zone. As explained in §3.1.3, this represents the embedding of a torus in $\mathbf{d}$ -space, shown in fig. 3.2.

The u -parameter has the effect of translating the torus by a fixed amount in the z -direction. From this we can see that the QWZ model only has non-zero Chern number for values of u between -2 and 2. Translating the torus by any more will move the origin completely outside. Thus the Chern number is

$$Q = \begin{cases} -1 & : -2 < u < 0, \\ 1 & : 0 < u < 2, \\ 0 & : \text{otherwise}, \end{cases}\tag{3.2.4}$$

not including the points at $u = -2, 0$ and 2 , at which the Chern number is undefined because the band gap closes and the material becomes conducting. Some example band structures for varying values of u are shown in fig. 3.3.

3.2.1 Strip Geometry and Edge Modes

According to the bulk-boundary correspondence, whenever the Chern number $\mathcal{C}$ is nonzero, the QWZ model must have gapless edge modes appearing. To provide an example, here we construct the model on strip geometry and provide an analytical expression for the edge modes.

We shall require that the system is periodic and uniform in the y -direction, but is allowed to be non-uniform and have open boundaries in the x -direction. We allow u to vary in the x -direction, so add a subscript to indicate position-dependence, $u \rightarrow u_{m_x}$. We can apply Bloch's theorem in the y -direction only, and our eigenstates take the following form,

$$|\psi\rangle = |k_y\rangle \otimes |v_{k_y}\rangle,\tag{3.2.5}$$

where $|k_y\rangle$ is a plane wave in the y direction and $|v_{k_y}\rangle$ is a $2 \times L_x$ element vector. The v -states

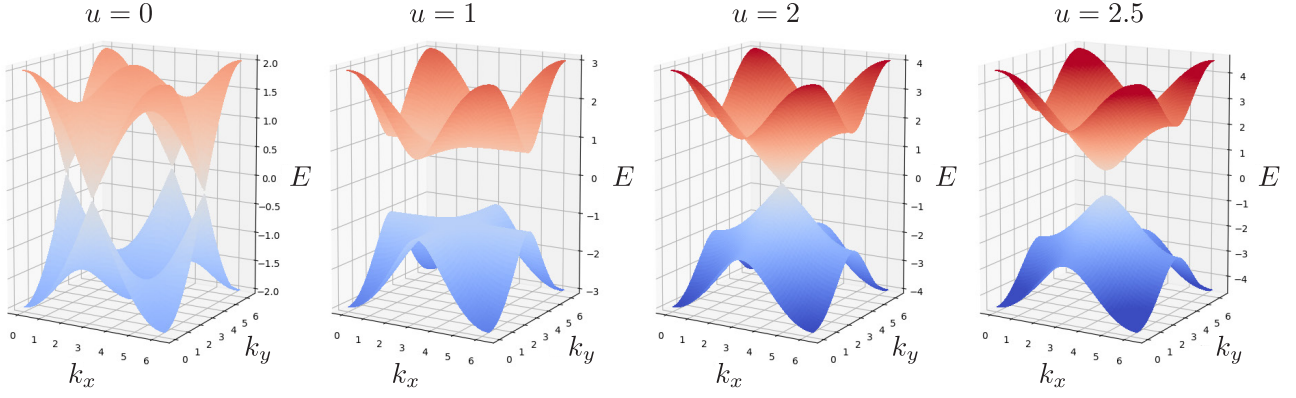


Figure 3.3: Four examples of the band structure of the QWZ model are shown, for varying u values between 0 and 2.5. At $u = 0$ and 2, the bands touch and Chern number is undefined. At $u = 1$, $\mathcal{C} = +1$, and at $u = 2.5$, $\mathcal{C} = 0$.

are solutions of the semi-bulk Hamiltonian

$$\begin{aligned} \hat{H}(k_y) = & \sum_{m_x} |m_x + 1\rangle \langle m_x| \otimes \frac{\sigma_z + i\sigma_x}{2} + h.c. \\ & + \sum_{m_x} |m_x\rangle \langle m_x| \otimes [(u_{m_x} + \cos k_y) \sigma_z + \sin k_y \sigma_y]. \end{aligned} \quad (3.2.6)$$

We have effectively reduced our system from a lattice of sites in x and y , to a set of chain models in x , that are each labelled by a value of k_y . The band structure for this system in both periodic boundaries and strip geometry is shown in fig. 3.4. Note that edge states are only present for certain values of k_y , which can be seen in figure 3.4, where the edge states peel away from the bulk in a region around $k_y = \pi$.

The strip Hamiltonian can be solved by proposing modes that decay exponentially away from the boundary. Thus, a left-edge state has the form

$$|\psi_{k_y}^{\text{left}}\rangle = \sqrt{1 - \xi_l^2} \sum_{m=1}^{L_x} \xi_l^{m-1} |m\rangle \otimes \begin{pmatrix} 1 \\ -i \end{pmatrix} \quad (3.2.7)$$

where

$$\xi_l = -u - \cos k_y. \quad (3.2.8)$$

This state has energy

$$E_l = -\sin k_y. \quad (3.2.9)$$

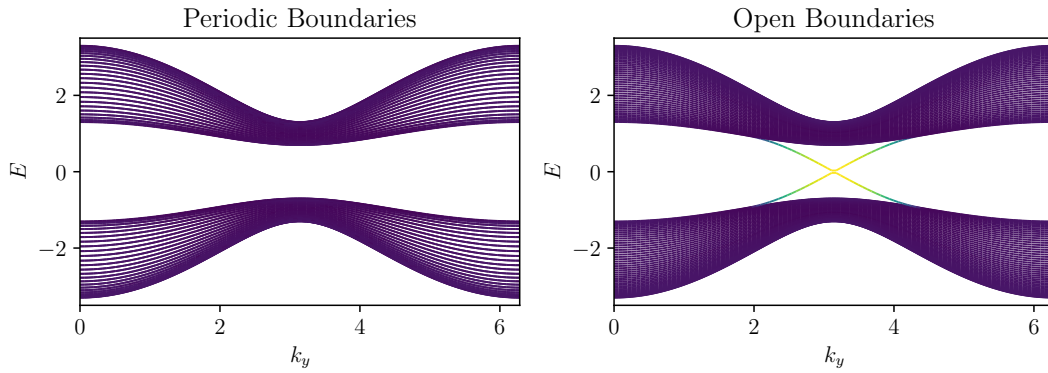


Figure 3.4: An example band structure for a QWZ model with $u = 1.3$ and $L_x = 30$. States are coloured according to their inverse participation ratio, indicating their relative degree of localisation. **(Left)** The example for periodic boundary conditions, showing two well-separated bands. **(Right)** The case for open boundary conditions. Two edge states are shown leaving the bulk and crossing the gap, with the descending state being a left edge state, and the ascending state being a right edge state.

Similarly, the right edge state is determined by

$$|\psi_{k_y}^{\text{right}}\rangle = \sqrt{1 - \xi_r^2} \sum_{m=1}^{L_x} \xi_r^{L_x - m} |m\rangle \otimes \begin{pmatrix} 1 \\ i \end{pmatrix} \quad (3.2.10)$$

$$\xi_r = -u - \cos k \quad (3.2.11)$$

$$E_r = \sin k. \quad (3.2.12)$$

These edge states are only valid solutions for k_y in the region

$$-1 < u + \cos k_y < 1. \quad (3.2.13)$$

These limits can be seen in fig. 3.4, where the edge states peel away from the bands to cross the energy gap. These expressions are only strictly valid in the limit of a large system. since we have made the assumption the edge states can be calculated by taking into consideration only one edge and ignoring the effect of the other. The edge states decay exponentially through the system, so we are relying on the assumption that the edge state has effectively decayed to nothing before reaching the other side. When this condition is relaxed the edge states are able to couple to one another. This has a two effects, the first being that at the crossing of edge states, our energy eigenvalues become positive and negative superposition states of both edge states. The other is that the left state coupling to the right state shifts their energy levels producing an avoided crossing such that no states cross $E = 0$.

3.2.2 The Amorphous QWZ Model

We now look at extending the QWZ model beyond a grid to arbitrary amorphous lattices. This is done by defining a smooth continuation of the QWZ model to allow for edges that can point in an arbitrary direction. Given an arbitrary lattice, we first require that every vertex hosts two states, and the on-site energy term containing $u\sigma_z$ is identical as in eqn 5.3.1. To construct an

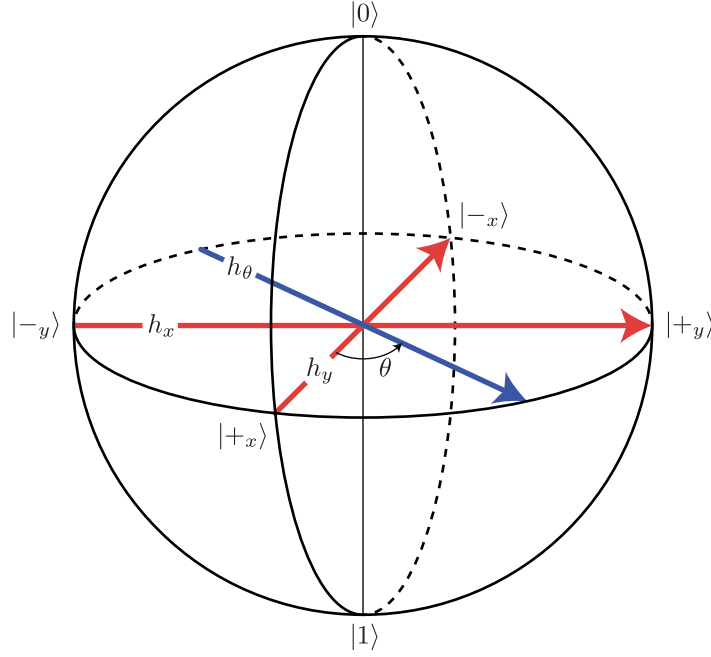


Figure 3.5: The Bloch sphere, showing the positions of the eigenstates of the three Pauli matrices. The action of the hopping matrices is shown in red, where the h_x and h_y operators act as creation operators on the y and x bases respectively. The action of our arbitrary angle hopping operator is shown in blue, acting as a creation operator between a pair of orthogonal states that lie on the x - y plane.

amorphous equivalent of the hopping terms, let us first look at the hopping in the QWZ model more carefully. We have two hopping matrices, one that hops a state in the x -direction, h_x , and one that hops in the y -direction, h_y .

$$h_x = \frac{\sigma_z + i\sigma_x}{2} = \frac{1}{2} \begin{pmatrix} 1 & i \\ i & 1 \end{pmatrix} \quad (3.2.14)$$

$$h_y = \frac{\sigma_z + i\sigma_y}{2} = \frac{1}{2} \begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix}. \quad (3.2.15)$$

These operators can be viewed as creation operators on different bases, h_x acts on the basis of σ_y , and h_y acts on the basis of σ_x . Viewed in the Bloch sphere, these operators hop the state across the sphere in the x or y direction.

In our amorphous lattice, hopping operators can act across an edge connecting two sites that points in an arbitrary direction. Thus, rather than trying to assign edges with h_x or h_y operators, let us define an arbitrary-angle hopping matrix h_θ , that hops in a direction θ over the x - y plane of the Bloch sphere,

$$h_\theta = |+\theta\rangle \langle -\theta| \quad (3.2.16)$$

$$= \frac{1}{2} \begin{pmatrix} 1 & -e^{-i\theta} \\ e^{i\theta} & -1 \end{pmatrix}. \quad (3.2.17)$$

Thus, the Hamiltonian we use for the amorphous system is given by

$$\begin{aligned}\hat{H} = & \sum_{i,j} |\mathbf{r}_i\rangle \langle \mathbf{r}_j| \otimes h_{\theta_{i,j}} + h.c. \\ & + \sum_i |\mathbf{r}_i\rangle \langle \mathbf{r}_i| \otimes u_i \sigma^z,\end{aligned}\tag{3.2.18}$$

where $\theta_{i,j}$ is the angle between the edge connecting $\mathbf{r}_i$ and $\mathbf{r}_j$ and the x axis.

As before, the parameter u controls whether the system is in a topological or trivial state. The exact dependence of the system on u depends strongly on the lattice geometry, however there is generally a region around which the system has nonzero topological character. In the absence of a momentum-space description, the system's topological character can be seen indirectly by comparing the spectrum in open and periodic boundary conditions. Generally one expects the presence of a gap in periodic boundaries that becomes populated with edge modes in open boundaries. Two examples are shown in fig. 3.6.

3.2.3 Topology of the Amorphous QWZ Model

OPTIONAL – you could add a section here on the low- k theory of the amorphous QWZ model if you wanted - would allow you to make some predictions for the Chern number as you can find the berry phase change around the low k band touching. Maybe its a bit overkill - the thesis looks like its gonna be kinda long anyway.

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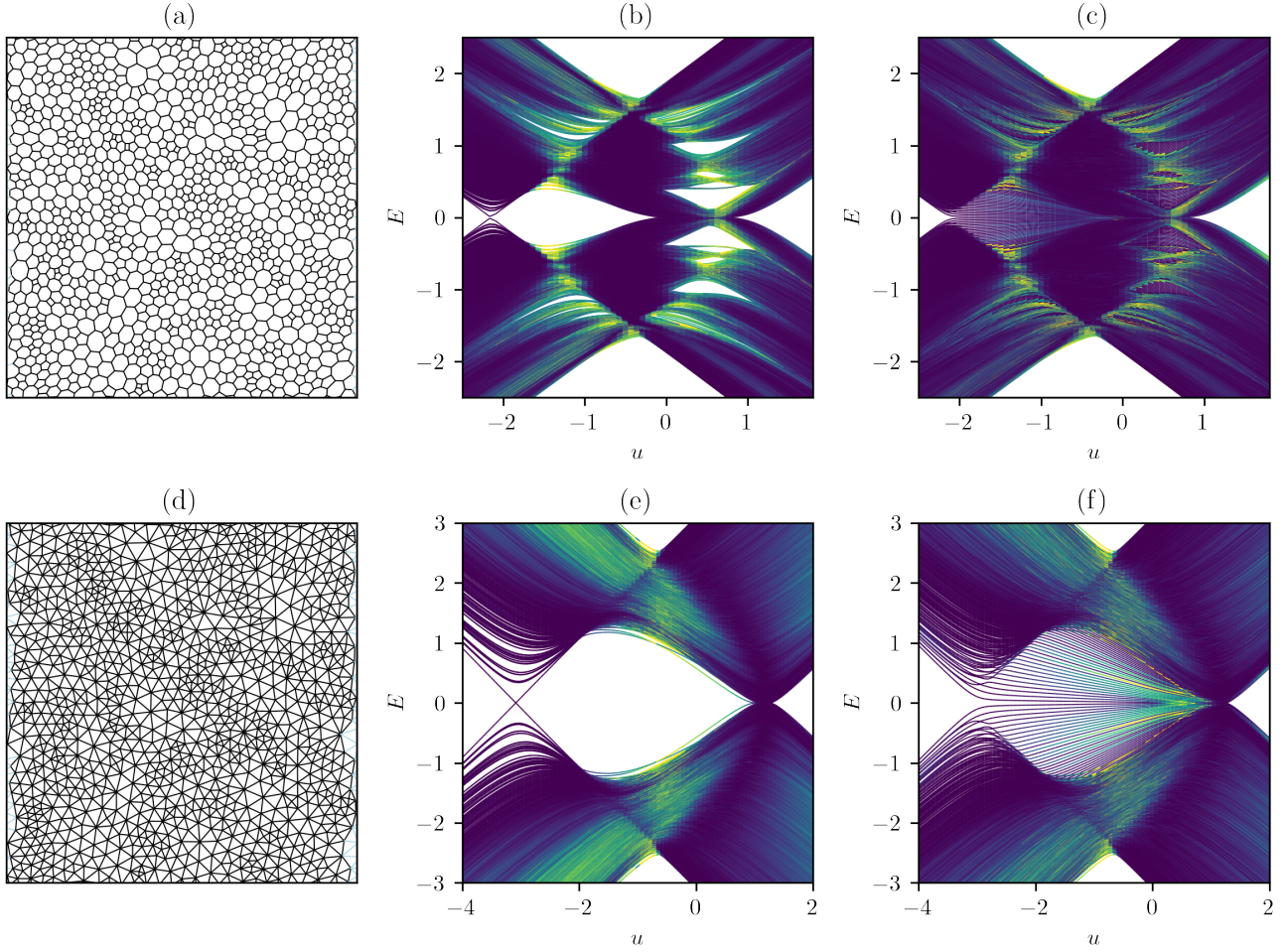


Figure 3.6: Plots showing the spectra of the amorphous QWZ model for two different example lattices. States are coloured according to their inverse participation ratio. **(a)** A trivalent lattice generated from a Voronoi decomposition of a set of random points, followed by five iterations of Lloyd's algorithm to regularise bond lengths. **(b)** The spectrum of the QWZ model on the voronised lattice as a function of u . Several gaps are present at different values of u and E . **(c)** The spectrum of the same model as (b), but in open boundary conditions. Here we can see that all gaps for $u \in [-1.5, 1]$ are now filled with edge modes, indicating that the system is in a topological phase. **(d)** An example of a lattice forming a Delaunay triangulation, constructed as the dual of the lattice in (a). **(e)** The spectrum of the QWZ model on the Delaunay lattice as a function of u showing one single large gapped region. **(f)** The spectrum of the same model as (e), but in open boundary conditions. Here we can see that the gaps is now filled with edge modes, again indicating that here the system is in a topological phase.

Chapter 4

Local Markers

The methods outlined in Chapter 3 have proved extremely effective for characterising electronic band structure [1–4] and even found use in describing ocean waves [5] coupled mechanical systems [6–8], as well as providing the foundations for the modern theory of polarisation [9–11]. However the definition we provided for the Chern number suffers from some important limitations. We are forced to work in Fourier space in order to construct the Berry curvature. Fourier space is only well-defined when the system in question has perfectly uniform translational symmetry, as well as living in either periodic or infinite boundary conditions. This is a serious problem – every material we encounter in the real world will fail to satisfy both of these conditions! Furthermore, it is not possible to apply any of these methods to quantum systems with interactions, or to systems out of equilibrium.

One potential solution to these problems has been the development of real space methods for calculating the Chern number. There are a litany of proposals, such as Kitaev’s Chern marker [8, 12], the Bott Index [13–16], the Chern marker [17, 18], the localizer index [19] and most recently structural spillage [20]. In all of these cases, the aim is to re-express the definition of Chern number in terms of a set of real-space operators which are local, and thus do not rely on the notion of Fourier space to make sense. Such a quantity should be equivalent to the Chern number when calculated in a translationally symmetric context; however would still be possible in cases where these conditions have been relaxed, and the notion of k -space is gone. This would afford the researcher tools with which to comment on the topological characteristics of materials that had previously been out of reach, such as amorphous crystals [21], quasi-periodic lattices [22] and systems out of equilibrium [14, 23–26] and even strongly interacting systems [27].

In this section we start by introducing three such quantities, the Kitaev Chern number, the Bott index and the Chern marker, giving their definitions as well as explaining how they are related.

add a bit more summary of wot u writing

4.1 Kitaev’s Local Marker

The majority of accounts of local markers tend to focus on the development of the Bott index and the Chern marker. However, the first construction of a spatial Chern number can, along with the rest of modern physics, be found in the appendices of a seminal work by Kitaev [12]. Thus, we shall start our discussion with an explanation of Kitaev’s local marker.

4.2 The Bott Index

We will work with a lattice model, where unit cells are indexed by a vector of the form (m_x, m_y) . The lattice has size (L_x, L_y) , and we have set periodic boundary conditions. Sites are indexed by

$$|\mathbf{m}\alpha\rangle = |\mathbf{m}\rangle \otimes |\alpha\rangle, \quad (4.2.1)$$

where $\mathbf{m}$ is a vector labelling the unit cell and α selects a site within the unit cell. We shall use a Hamiltonian that has translational symmetry, so we may apply Bloch's theorem to obtain the form of the energy eigenstates,

$$\hat{H} = \sum_{\mathbf{k}} |\mathbf{k}\rangle \langle \mathbf{k}| \otimes \hat{H}(\mathbf{k}) \quad (4.2.2)$$

$$|\psi_{\mathbf{k},n}\rangle = |\mathbf{k}\rangle \otimes |u_{\mathbf{k},n}\rangle, \quad (4.2.3)$$

where $|u_{\mathbf{k},n}\rangle$ are the eigenstates of $\hat{H}(\mathbf{k})$ and the momentum states are defined as in §??,

$$|\mathbf{k}\rangle = \frac{1}{\sqrt{L_x L_y}} \sum_{\mathbf{m}} e^{i\mathbf{m}\cdot\mathbf{k}} |\mathbf{m}\rangle, \quad (4.2.4)$$

and we assume that the system has a total of η bands, so $n \in \{1, \dots, \eta\}$. Henceforth we will assume that only the n^{th} band is occupied, so we can construct a projector onto the occupied subspace, $\hat{P}$, and its complement, $\hat{Q}$.

$$\hat{P} = \sum_{\mathbf{k}} |\psi_{\mathbf{k},n}\rangle \langle \psi_{\mathbf{k},n}| \quad (4.2.5)$$

$$\hat{Q} = \sum_{\mathbf{k}, m \neq n} |\psi_{\mathbf{k},m}\rangle \langle \psi_{\mathbf{k},m}|. \quad (4.2.6)$$

We also introduce a pair of operators $e^{i\delta_x \hat{x}}$ and $e^{i\delta_y \hat{y}}$, with

$$\delta_x = \frac{2\pi}{L_x}, \quad \delta_y = \frac{2\pi}{L_y}. \quad (4.2.7)$$

In order to understand the effect of these operators, we consider their effect on the plane wave states

$$e^{i\delta_x \hat{x}} |\mathbf{k}\rangle = \frac{1}{\sqrt{L_x L_y}} \sum_{\mathbf{m}} e^{i\delta_x \hat{x}} e^{i\mathbf{m}\cdot\mathbf{k}} |\mathbf{m}\rangle \quad (4.2.8)$$

$$= \frac{1}{\sqrt{L_x L_y}} \sum_{\mathbf{m}} e^{i\mathbf{m}\cdot(\mathbf{k}+\boldsymbol{\delta}_x)} |\mathbf{m}\rangle \quad (4.2.9)$$

$$= |\mathbf{k} + \boldsymbol{\delta}_x\rangle, \quad (4.2.10)$$

similarly

$$e^{i\delta_y \hat{y}} |\mathbf{k}\rangle = |\mathbf{k} + \boldsymbol{\delta}_y\rangle \quad (4.2.11)$$

Here the symbols δ_x and δ_y represents a vector translation in $\mathbf{k}$ -space,

$$\delta_x = \begin{pmatrix} \delta_x \\ 0 \end{pmatrix}, \quad \delta_y = \begin{pmatrix} 0 \\ \delta_y \end{pmatrix}. \quad (4.2.12)$$

Thus, these operators effect a translation in $\mathbf{k}$ space by the minimum step in the reciprocal lattice.

Now let's look at the effect of these operators on the projector $\hat{P}$,

$$e^{i\delta_x \hat{x}} \hat{P} e^{-i\delta_x \hat{x}} = \sum_{\mathbf{k}} e^{i\delta_x \hat{x}} |\mathbf{k}\rangle \langle \mathbf{k}| e^{-i\delta_x \hat{x}} \otimes |u_{\mathbf{k},n}\rangle \langle u_{\mathbf{k},n}| \quad (4.2.13)$$

$$= \sum_{\mathbf{k}} |\mathbf{k} + \delta_x\rangle \langle \mathbf{k} + \delta_x| \otimes |u_{\mathbf{k},n}\rangle \langle u_{\mathbf{k},n}| \quad (4.2.14)$$

$$= \sum_{\mathbf{k}} |\mathbf{k}\rangle \langle \mathbf{k}| \otimes |u_{\mathbf{k}-\delta_x,n}\rangle \langle u_{\mathbf{k}-\delta_x,n}|, \quad (4.2.15)$$

where the last line follows from a redefinition of $\mathbf{k}$. This is allowed as we are summing over the whole Brillouin zone. For the sake of clarity later on, we will re-express the density operator $|u_{\mathbf{k},n}\rangle \langle u_{\mathbf{k},n}|$ in the following form

$$|u_{\mathbf{k},n}\rangle \langle u_{\mathbf{k},n}| = \hat{u}_{\mathbf{k},n}, \quad (4.2.16)$$

so we get

$$e^{i\delta_x \hat{x}} \hat{P} e^{-i\delta_x \hat{x}} = \sum_{\mathbf{k}} |\mathbf{k}\rangle \langle \mathbf{k}| \otimes \hat{u}_{\mathbf{k}-\delta_x,n}. \quad (4.2.17)$$

Now we have enough tools to introduce the definition of the Bott index!

$$B(\mathbf{A}) = -\frac{N}{2\pi} \sum_{\alpha} \Im \langle \mathbf{A}\alpha | \log [\hat{U}\hat{V}\hat{U}^\dagger\hat{V}^\dagger + \hat{Q}] | \mathbf{A}\alpha \rangle \quad (4.2.18)$$

Here N denotes the total number of unit cells in the system and the log is a standard matrix logarithm, defined as

$$\log(M) = \sum_{k=1}^{\infty} (-1)^{k+1} \frac{(M - \mathbb{1})^k}{k}. \quad (4.2.19)$$

The Bott index is a local indicator, $\mathbf{A}$ is a vector that selects which unit cell in the system we are looking at, and

$$\hat{U} = \hat{P} e^{i\delta_x \hat{x}} \hat{P} \quad (4.2.20)$$

$$\hat{V} = \hat{P} e^{i\delta_y \hat{y}} \hat{P}. \quad (4.2.21)$$

To make sense of this expression, let us focus on the $\hat{U}\hat{V}\hat{U}^\dagger\hat{V}^\dagger$ part only, writing this out explicitly gives us

$$\hat{U}\hat{V}\hat{U}^\dagger\hat{V}^\dagger = \hat{P} e^{i\delta_x \hat{x}} \hat{P} e^{i\delta_y \hat{y}} \hat{P} e^{-i\delta_x \hat{x}} \hat{P} e^{-i\delta_y \hat{y}} \hat{P}. \quad (4.2.22)$$

We now add in the identity in the form $e^{i\delta_x \hat{x}} e^{-i\delta_x \hat{x}}$, as well as the corresponding expression for y ,

and regroup the equation, taking advantage of the fact that $\hat{x}$ and $\hat{y}$ commute, to get

$$\hat{U}\hat{V}\hat{U}^\dagger\hat{V}^\dagger = \hat{P} \left[e^{i\delta_x\hat{x}} \hat{P} e^{-i\delta_x\hat{x}} \right] \left[e^{i\delta_y\hat{y}} e^{i\delta_x\hat{x}} \hat{P} e^{-i\delta_y\hat{y}} e^{-i\delta_x\hat{x}} \right] \left[e^{i\delta_y\hat{y}} \hat{P} e^{-i\delta_y\hat{y}} \right] \hat{P}. \quad (4.2.23)$$

Using eqn. 4.2.17 we can now rewrite this in terms of Bloch states to get

$$\hat{U}\hat{V}\hat{U}^\dagger\hat{V}^\dagger = \sum_{\mathbf{k}} |\mathbf{k}\rangle \langle \mathbf{k}| \otimes \hat{u}_{\mathbf{k},n} \hat{u}_{\mathbf{k}-\delta_x,n} \hat{u}_{\mathbf{k}-\delta_x-\delta_y,n} \hat{u}_{\mathbf{k}-\delta_y,n} \hat{u}_{\mathbf{k},n}. \quad (4.2.24)$$

Since the first and last $\hat{u}$ density operator in this expression are identical, we can rewrite this in terms of a trace to get

$$\hat{U}\hat{V}\hat{U}^\dagger\hat{V}^\dagger = \sum_{\mathbf{k}} |\mathbf{k}\rangle \langle \mathbf{k}| \otimes |u_{\mathbf{k},n}\rangle \langle u_{\mathbf{k},n}| \cdot \text{Tr} \left[\hat{u}_{\mathbf{k},n} \hat{u}_{\mathbf{k}-\delta_x,n} \hat{u}_{\mathbf{k}-\delta_x-\delta_y,n} \hat{u}_{\mathbf{k}-\delta_y,n} \right]. \quad (4.2.25)$$

If we compare this to the definition of Berry flux (eqn. 3.1.9), we see that this reduces to

$$\hat{U}\hat{V}\hat{U}^\dagger\hat{V}^\dagger = \sum_{\mathbf{k}} |\mathbf{k}\rangle \langle \mathbf{k}| \otimes |u_{\mathbf{k},n}\rangle \langle u_{\mathbf{k},n}| \cdot r_{\mathbf{k}} e^{-iF_{\mathbf{k}} - \delta_x - \delta_y}, \quad (4.2.26)$$

where $r_{\mathbf{k}}$ is just some real number that we don't care about, arising from the u states not being perfectly parallel to one another. This is promising, remember we are aiming to get a sum of $F_{\mathbf{k}}$ over the whole Brillouin zone. Thus in order to extract the term in the exponential we would like to take a log. However, we cannot simply log the expression in the form we have here. This is because it only has support on the occupied subspace,

$$\hat{U}\hat{V}\hat{U}^\dagger\hat{V}^\dagger \sim \begin{pmatrix} M & 0 \\ 0 & 0 \end{pmatrix}, \quad (4.2.27)$$

where M just indicates the presence of a nonzero operator. The presence of a null space means a log of this matrix will diverge. To remedy this, we add in the $\hat{Q}$ projector,

$$\hat{U}\hat{V}\hat{U}^\dagger\hat{V}^\dagger + \hat{Q} \sim \begin{pmatrix} M & 0 \\ 0 & \mathbb{1} \end{pmatrix}. \quad (4.2.28)$$

Now we can safely take the log of this matrix, getting the following result

$$\log \left[\hat{U}\hat{V}\hat{U}^\dagger\hat{V}^\dagger + \hat{Q} \right] = \sum_{\mathbf{k}} |\mathbf{k}\rangle \langle \mathbf{k}| \otimes |u_{\mathbf{k},n}\rangle \langle u_{\mathbf{k},n}| \cdot (\log(r_{\mathbf{k}}) - iF_{\mathbf{k}} - \delta_x - \delta_y), \quad (4.2.29)$$

Next we trace out the internal degree of freedom and squeeze this expression between a pair of position states $|\mathbf{A}\rangle$, selecting some arbitrary unit cell in our system.

$$\sum_{\alpha} \langle \mathbf{A}\alpha | \log \left[\hat{U}\hat{V}\hat{U}^\dagger\hat{V}^\dagger + \hat{Q} \right] | \mathbf{A}\alpha \rangle = \sum_{\mathbf{k}} |\langle \mathbf{A} | \mathbf{k} \rangle|^2 \cdot \langle u_{\mathbf{k},n} | u_{\mathbf{k},n} \rangle \cdot (\log(r_{\mathbf{k}}) - iF_{\mathbf{k}} - \delta_x - \delta_y). \quad (4.2.30)$$

We can show that $|\langle \mathbf{A} | \mathbf{k} \rangle|^2 = \frac{1}{N}$ and $\langle u_{\mathbf{k},n} | u_{\mathbf{k},n} \rangle = 1$. All that is left to do is take the imaginary

part,

$$\sum_{\alpha} \Im \langle \mathbf{A}\alpha | \log [\hat{U}\hat{V}\hat{U}^{\dagger}\hat{V}^{\dagger} + \hat{Q}] | \mathbf{A}\alpha \rangle = \frac{1}{N} \sum_{\mathbf{k}} -F_{\mathbf{k}-\delta_x-\delta_y}. \quad (4.2.31)$$

Now, redefining $\mathbf{k}$ and multiplying by $-\frac{N}{2\pi}$ gives us

$$-\frac{N}{2\pi} \sum_{\alpha} \Im \langle \mathbf{A}\alpha | \log [\hat{U}\hat{V}\hat{U}^{\dagger}\hat{V}^{\dagger} + \hat{Q}] | \mathbf{A}\alpha \rangle = \frac{1}{2\pi} \sum_{\mathbf{k}} F_{\mathbf{k}}. \quad (4.2.32)$$

By comparing with eqn. 3.1.12, we can see that we have arrived at an expression for the Chern number!

$$\mathcal{Q} = -\frac{N}{2\pi} \sum_{\alpha} \Im \langle \mathbf{A}\alpha | \log [\hat{U}\hat{V}\hat{U}^{\dagger}\hat{V}^{\dagger} + \hat{Q}] | \mathbf{A}\alpha \rangle. \quad (4.2.33)$$

Thus we have shown that in the case of a translationally symmetric system with periodic boundaries, the Bott index evaluated at any point in our material always exactly calculates the Chern number. Before we show some example calculations, let us introduce one more quantity.

4.3 Chern Marker

The Chern marker was first introduced in [17], however we will not restate the derivation that they used. Instead we will justify the Chern marker by showing that it can be obtained from the Bott index in the limit of large system size.

Our starting point is the expression for Bott index, eqn. 4.2.18. We start by multiplying $\hat{U}$ and $\hat{V}$ by complex phases $e^{-i\delta_x C_x}$ and $e^{-i\delta_y C_y}$. This has no effect on the value of the Bott index, since the phases have opposite sign in $\hat{U}^{\dagger}$ and $\hat{V}^{\dagger}$.

$$\hat{U} \rightarrow \hat{P} e^{i\delta_x(\hat{x}-C_x)} \hat{P} \quad (4.3.1)$$

$$\hat{V} \rightarrow \hat{P} e^{i\delta_y(\hat{y}-C_y)} \hat{P}. \quad (4.3.2)$$

As the system size becomes large, the factors δ_x and δ_y will become infinitesimally small. Around the neighbourhood of the unit cell at position (C_x, C_y) , the term in the exponential will be small so we can Taylor expand it,

$$\hat{U} = \hat{P} + i\delta_x \hat{P} \hat{x}_C \hat{P} + O(\hat{x}_C^2) \quad (4.3.3)$$

$$\hat{V} = \hat{P} + i\delta_y \hat{P} \hat{y}_C \hat{P} + O(\hat{y}_C^2) \quad (4.3.4)$$

where we have relabelled $\hat{x} - C_x = \hat{x}_C$ and $\hat{y} - C_y = \hat{y}_C$. Using these expressions, we can re-express the operator $\hat{U}\hat{V}\hat{U}^{\dagger}\hat{V}^{\dagger}$ up to second order in δ as¹

$$\hat{U}\hat{V}\hat{U}^{\dagger}\hat{V}^{\dagger} = \hat{P} - \delta_x \delta_y \left(\hat{P} \hat{x}_C \hat{P} \hat{y}_C \hat{P} - \hat{P} \hat{y}_C \hat{P} \hat{x}_C \hat{P} \right) + O(\delta_x^2, \delta_y^2) \quad (4.3.5)$$

¹You might have noticed that we have omitted the δ_x^2 and δ_y^2 terms. This is justified because those terms involve only hermitian operators, such as $\hat{P} \hat{x}_C \hat{P} \hat{x}_C \hat{P}$, and these give no contribution when we take the imaginary part later.

Now we use the identities

$$\hat{P} + \hat{Q} = \mathbb{1} \quad (4.3.6)$$

$$\log [\mathbb{1} + \delta M] \simeq M, \text{ for } \delta \text{ small} \quad (4.3.7)$$

to show that

$$\log [\hat{U}\hat{V}\hat{U}^\dagger\hat{V}^\dagger + Q] \simeq -\delta_x\delta_y [\hat{P}\hat{x}_C\hat{P}, \hat{P}\hat{y}_C\hat{P}] \quad (4.3.8)$$

We can now drop the constant shifts in $\hat{x}_C$ and $\hat{y}_C$, since these vanish due to the commutator. Inserting this into the expression for Bott index, we arrive at the form of the Chern marker,

$$C(\mathbf{A}) = -2\pi i \sum_{\alpha} \langle \mathbf{A}\alpha | [\hat{P}\hat{x}\hat{P}, \hat{P}\hat{y}\hat{P}] | \mathbf{A}\alpha \rangle. \quad (4.3.9)$$

We have skipped taking the imaginary part by noticing that the commutator is anti-Hermitian, so all of its eigenvalues are imaginary anyway, allowing us to make the substitution $\Im \rightarrow -i$.

This calculation comes with one important caveat. The Chern marker has an extra property that the Bott index lacks. If the system in question has finite size, then the sum of the Chern number over the whole system vanishes. This is because the sum of $C(\mathbf{A})$ over the system is given by

$$\sum_{\mathbf{A}} C(\mathbf{A}) = -2\pi i \text{Tr} [\hat{P}\hat{x}\hat{P}, \hat{P}\hat{y}\hat{P}], \quad (4.3.10)$$

and the trace of a commutator is always zero in a finite Hilbert space.

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Chapter 5

The Crosshair Marker

Since the initial discovery of quantised Hall conductance [1–3] and throughout the subsequent decades spent exploring novel topological phases of matter [4, 5], the Chern number has been one of the central tools for understanding condensed matter systems outside of the Landau symmetry-breaking framework [6]. In the study of non-interacting Chern insulators, different Hamiltonians can be classified according to the Chern number of their bands. Two Hamiltonians with differing Chern numbers cannot be smoothly deformed into one another without crossing a point where the system becomes conductive, and systems with nonzero Chern numbers will always have conducting edge modes at their boundaries [7]. The Chern number has also been extensively used outside of topological insulators, an example being the Kitaev spin-liquid model where it indicates whether the system has Abelian or non-Abelian anyons [8].

A defining characteristic of quantum Hall physics is the robustness of the edge states and Hall conductance to impurities in the sample. Despite this, the Chern number – the central quantity in the TKNN invariant determining the Hall conductance [3] – cannot be calculated in a disordered material. The Chern number is calculated for a single band, and depends on the structure of the Brillouin zone. If a material lacks translational symmetry, it is not possible to apply Bloch’s theorem, crystal momentum is not a good quantum number and we have no concept of momentum space. Disorder-resistant methods exist [9, 10], however they compute a global Chern number, and cannot capture the local properties of materials with compound structure. This tension, that our tools for describing disorder-resistant physics are themselves undermined by disorder, suggests that our mathematical framework is incomplete.

One possible solution to this problem is the development of local Chern markers. These attempt to translate the Chern number into a mathematical language that is resolved in real space, ensuring it is well-defined when the system has no translational symmetry. Several candidates for local markers exist, such as the Chern marker [11], the Bott index [12], and the Chern number defined by Kitaev in appendix C of [8]. Each of these markers has been derived by finding ways of re-expressing the Chern number in terms of quantities that can be evaluated in real space. This ensures that in an ideal uniform material with no boundaries – where the Chern number can be calculated directly – the marker should exactly equal the Chern number. Despite satisfying this requirement, existing markers suffer from two main drawbacks. Firstly, they display unexplained behaviour when taken out of the context where the Chern number is already well-defined. The Chern marker and Bott index have sharp drops around the boundary of any topological region [13], and Kitaev’s local Chern number vanishes in non-infinite systems. Secondly, all markers were developed from a mathematical restatement of the Chern number, and the resulting expression does not obviously correspond to any physical quantity. Thus, it is

not clear at all how one might attempt to measure it in a real system, or even if there is a way to connect them to an observable at all – although some attempts have been made [14–16].

We present a derivation for a local marker that starts from physical grounds, defined as a localised version of the Hall conductivity [17]. Local current is measured around the position $\mathbf{R}$ of a crosshair placed in the bulk of the material, motivating the name ‘crosshair marker’. The derived quantity is close to the local Chern marker defined by Kitaev [8], given by

$$C(\mathbf{R}) = 4\pi \Im \text{Tr}_{\text{Bulk}} (P \vartheta_{R_x} P \vartheta_{R_y} P), \quad (5.0.1)$$

where P is a projector onto the occupied band and ϑ is a step function at $\mathbf{R}$ in the x or y directions. The trace is over a region around $\mathbf{R}$ in the bulk. Since the marker has a straightforward interpretation, its behaviour around edges of the system can be understood intuitively in terms of current induced in the material. Furthermore, when evaluated in a disordered system, the crosshair marker is almost exactly quantised provided that it is measured sufficiently far from any edge modes. In contrast, the Bott index and Chern marker are not strictly quantised in the presence of disorder. We also find a connection between the crosshair marker and the Chern marker. Summing over all possible positions of the crosshair itself, we exactly recover the Chern marker, allowing for the physical interpretation of the crosshair marker to be extended to the Chern marker.

The paper is structured as follows: In §5.1, we derive the necessary pre-requisites to understand the marker. These are threefold, in §5.1.1 we describe the formalism for modelling the effect of electric fields on our system, in §5.1.2 we define a set of current operators on a lattice, and in §5.1.3 we describe Kato’s formalism for adiabatic quantum evolution based on [18]. In §5.2 we use these concepts to derive the crosshair marker and discuss its connection to the Chern marker. Finally, in §5.3 we give examples of the crosshair marker in the Qi-Wu-Zhang (QWZ) model with spatially-varying parameters, as well as an extension of the QWZ Hamiltonian to amorphous lattices. These serve as a testing ground for extending the Chern number to systems that are inaccessible to the conventional momentum space calculations.

5.1 Preliminaries

We work with a general non-interacting Hamiltonian on a two-dimensional tight-binding lattice. The system consists of set of N sites, with positions $\mathbf{r}_i$ arranged on either a regular or amorphous lattice. Each site has an internal degree of freedom hosting η states. The Hamiltonian can be written in the form

$$H = \sum_{i,j} |\mathbf{r}_i\rangle \langle \mathbf{r}_j| \otimes H_{i,j}, \quad (5.1.1)$$

where $H_{i,j} = H_{j,i}^\dagger$ acts on the internal degrees of freedom. Here we will only work with systems in open boundary conditions.

5.1.1 Electric Fields

In §5.2 we examine the current generated by step-function electric potentials. Thus, here we precisely define the form of the electric field and potential, and study how they are represented on the lattice.

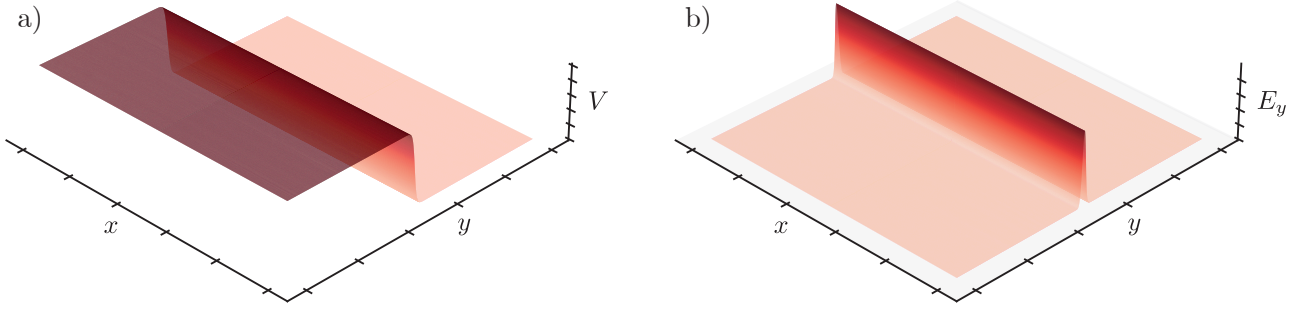


Figure 5.1: a) The electric potential used to generate our electric field. b) The electric field, with only the nonzero y -component is shown.

We wish to model the effect of raising the potential of one half of the system with respect to the other, resulting in an electric field across the boundary separating the two regions. Without loss of generality, let us consider raising the potential of the region below a dividing line at position $y = R_y$ with respect to the region above the line. In the continuum, this corresponds to an electric potential of the form $V(\mathbf{r}) = -V_0\theta(y - R_y)$, where θ is a Heaviside step function. The electric field acts over the line at $y = R_y$, as shown in figure 5.1,

$$\mathbf{E}(\mathbf{r}) = V_0\delta(y - R_y)\mathbf{e}_y, \quad (5.1.2)$$

where $\mathbf{e}_y$ is a unit vector in the y direction. We transform to a gauge with zero scalar potential, representing the electric field with a magnetic vector potential,

$$\mathbf{A}(\mathbf{r}, t) = -V_0t\delta(y - R_y)\mathbf{e}_y. \quad (5.1.3)$$

Making the assumption that V_0 is small, we shall use Peierls substitution to describe the effect of a slowly varying magnetic vector potential on our lattice system [19]. The translation operators are modified by a Peierls phase $|\mathbf{r}_j\rangle\langle\mathbf{r}_i| \rightarrow |\mathbf{r}_j\rangle\langle\mathbf{r}_i|e^{i\alpha(\mathbf{r}_i, \mathbf{r}_j)}$, given by

$$\alpha(\mathbf{r}_i, \mathbf{r}_j) = \int_{\mathbf{r}_j}^{\mathbf{r}_i} \mathbf{A}(\mathbf{r}) \cdot d\mathbf{r}. \quad (5.1.4)$$

Note that we work in natural units, setting the electronic charge $q = \hbar = 1$. Given the form of the magnetic vector potential, we can see that translation operators only pick up a phase if they cross the line $y = R_y$. This is illustrated in fig. 5.2. The Peierls phase can be expressed as

$$\alpha(\mathbf{r}_i, \mathbf{r}_j) = A(t) [\theta(y_i - R_y) - \theta(y_j - R_y)], \quad (5.1.5)$$

with $A(t) = V_0t$. Thus, the Hamiltonian is modified according to

$$H(A) = e^{iA(t)\vartheta_{R_y}} H e^{-iA(t)\vartheta_{R_y}}, \quad (5.1.6)$$

where we have defined the projector onto the half space above $y = R_y$ as

$$\vartheta_{R_y} = \sum_i \theta(y_i - R_y) |\mathbf{r}_i\rangle\langle\mathbf{r}_i|. \quad (5.1.7)$$

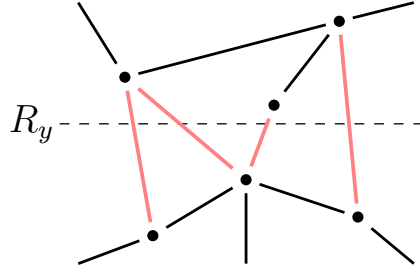


Figure 5.2: A section of amorphous lattice is shown. All hopping terms in the Hamiltonian that cross the line $y = R_y$ acquire a phase given by eqn. 5.1.5, these are depicted in red. Unaffected hopping terms are shown in black. Note that the phase is ill-defined when a site $\mathbf{r}_i$ lies on the line.

In the limit of small A , this can be expanded to first order,

$$H(A) = H - iA(t) [H, \vartheta_{R_y}]. \quad (5.1.8)$$

As we shall see in the next section, the change in the Hamiltonian is expressed in terms of a current operator for flow across the line $y = R_y$.

Note that for a static $A(t)$ in open boundaries, the A -dependence can be completely removed by a gauge transformation, so there will be no physical effect on our system. This is reflected by the fact that $\mathbf{E}$ depends on $\partial_t A$, so static A corresponds to zero electric field.

5.1.2 Current Operators

We construct an operator representing the flow of particles across a line bisecting the system. Again, without loss of generality, we will look at the flow of particles in the x -direction, across the vertical boundary at $x = R_x$. The number operator for particles to the right of this line is given by ϑ_{R_x} , defined analogously to eqn. 5.1.7. The time-evolution for the number of particles in this region is $\partial_t \langle \vartheta_{R_x} \rangle = i \langle [H, \vartheta_{R_x}] \rangle$. Thus, we can identify the operator representing flow of particles across the boundary as

$$J_{R_x} = i[H, \vartheta_{R_x}]. \quad (5.1.9)$$

This allows us to re-express eqn. 5.1.8 in the familiar form $H(A) = H - AJ_{R_y}$, where A is defined in equations 5.1.5.

The operator J_{R_x} evaluates the overall flow across the whole system, however in the following discussion we wish to separate the bulk and edge contributions. Thus let us decompose this operator into a sum of terms local to some point at $\mathbf{r}$, according to

$$J_{R_x}(\mathbf{r}) = \frac{1}{2} (\delta_{\mathbf{r}} J_{R_x} + J_{R_x} \delta_{\mathbf{r}}), \quad (5.1.10)$$

with $\delta_{\mathbf{r}} = |\mathbf{r}\rangle \langle \mathbf{r}|$. The two terms evaluate the contributions to J_{R_x} from hopping into site $\mathbf{r}$, and from hopping out of site $\mathbf{r}$ respectively. The factor of $1/2$ is included to ensure that $\sum_{\mathbf{r}} J_{R_x}(\mathbf{r}) = J_{R_x}$, since every pair of sites is counted twice. This decomposition is not unique – there are many ways one could express J_{R_x} as a sum over local terms. However, the final result

will be insensitive to particular method since we shall sum over all the bulk contributions, which will be well-separated from edge contributions.

5.1.3 Kato Dynamics

We will work in the adiabatic limit, with a Hamiltonian that changes over a long timescale T , where the local density of states in the bulk has an energy gap Δ . We define the projector onto the occupied states at $t = 0$ as $P_0 = \sum_{\text{occ}} |\psi_i(0)\rangle \langle \psi_i(0)|$, where $|\psi_i(0)\rangle$ is an eigenstate of H at $t = 0$. In general, time-evolution will result in a final state $P(t)$ that is related to the initial state by the Schrödinger equation. However, if the evolution is sufficiently slow (for $T \gg \Delta^{-1}$), we expect that P_0 will adiabatically evolve to the instantaneous projector, $P_I(t) = \sum_{\text{band}} |\psi_i(t)\rangle \langle \psi_i(t)|$, where $|\psi_i(t)\rangle$ are eigenstates of H at time t [20]. To ensure that our time-evolution is exactly adiabatic, we use a formalism based on that introduced in [18], although a modern description can be found in [20–22]. We sketch out the basics here. However for a detailed discussion, see appendix ??.

Rather than using H to generate time evolution, we define an adiabatic Hamiltonian, K , which satisfies the instantaneous Von-Neumann equation

$$\partial_t P_I(t) = -i[K, P_I], \quad (5.1.11)$$

where the change in P_I follows the instantaneous eigenstates of H . Using the identity $P\dot{P}P = 0$ (where dot denotes a time-derivative), we can verify that the equation is satisfied by K of the form

$$K(t) = i[\dot{P}_I, P_I]. \quad (5.1.12)$$

The full time evolution of the projector follows the instantaneous eigenstates in the limit of large $T \gg \Delta$. Thus, in the following discussion, where we are working in the adiabatic regime, we may use eqn. 5.1.11 to describe the change in P , and drop the subscript on P_I , since all our time evolution is adiabatic. Having defined the adiabatic Hamiltonian, let us examine the exact form of K for our applied electric field, as well as deriving an adiabatic equivalent of the current operator in eqn. 5.1.10.

Adiabatic EM Fields

The time-dependence of our Hamiltonian is given by eqn. 5.1.6. We work in open boundaries, so the operator $e^{iA(t)\vartheta_{R_y}}$ behaves as a straightforward unitary rotation on the eigenstates. Note that in periodic boundaries, this operator would enforce twisted boundary conditions, shifting both the eigenstates and eigenenergies of H . Thus, the time-dependence of the projector is

$$P(t) = e^{iA(t)\vartheta_{R_y}} P_0 e^{-iA(t)\vartheta_{R_y}}, \quad (5.1.13)$$

and the time-derivative is $\dot{P} = i\dot{A} [\vartheta_{R_y}, P]$. This can be inserted into definition 5.1.12 to arrive at the form of the adiabatic Hamiltonian,

$$K = \dot{A} (P\vartheta_{R_y}Q + Q\vartheta_{R_y}P), \quad (5.1.14)$$

where $Q = \mathbb{1} - P$ is the complement of the projector.

Adiabatic Current

The adiabatic current operator is derived following an analogous argument to that in §5.1.2. We find the change in particle number for the region $x > R_x$, given by ϑ_{R_x} . In this case, however, we use the adiabatic Von-Neumann equation 5.1.11 to express the expectation value, $\partial_t \langle \vartheta_{R_x} \rangle = i \langle [K, \vartheta_{R_x}] \rangle$. Thus, we can identify an adiabatic current operator

$$J_{R_x}^A = i[K, \vartheta_{R_x}]. \quad (5.1.15)$$

As before, we can localise this operator to extract only the current hopping to and from a position $\mathbf{r}$ according to

$$J_{R_x}^A(\mathbf{r}) = \frac{1}{2} \{ \delta_{\mathbf{r}}, J_{R_x}^A \}. \quad (5.1.16)$$

In appendix ?? we discuss carefully how the non-adiabatic expression for current relates to the adiabatic form as we approach the adiabatic limit.

5.2 Marker Derivation

Before we give the formal derivation, let us describe the physical intuition behind our topological marker, which can be understood as a localised version of the Hall conductivity. The standard Hall conductivity is measured by applying a uniform electric field to a 2D material and measuring the current induced perpendicular to the field. This current is quantised and corresponds to the Chern number of the Hamiltonian [2]. A disadvantage of this calculation is that it is necessarily global, with the electric field uniform over the system and the current measured across the whole sample. In a material with disorder or compound structure, the calculation gives no spatially-resolved information about its topological properties.

Our objective is to find an analogous quantity that can be spatially resolved, giving information about which parts of a complex material are responsible for topological behaviour. It is not obvious how one should localise the Hall conductivity: neither the electric field, nor the current operator can be easily localised to a point. Maxwell's laws prohibit one from creating an electric field at a single point only. Equally, on the lattice, it is difficult to write down a consistent expression for current at a point.

Despite that fact that neither electric field nor current can individually be made local to a point, it is possible to localise the overall Hall conductance. This is because both field and current can be localised to a line. By partitioning the system, and raising the electric potential of one part, we induce an electric field acting across the dividing line. Similarly, given a partition, by measuring the transfer of electrons from one part to the other, we are able to determine the current across the dividing line.

The route to a local Hall current is to partition the system along two perpendicular dividing lines, as shown in fig. 5.3. An electric field is induced between the top and bottom parts of the system by uniformly raising the voltage of the lower part. This will induce a vertical current acting across the dividing line. When the material has nonzero Chern number, a Hall current will also be induced from left to right. If the system is insulating (i.e. gapped) we expect that all current should be localised to the region around the horizontal line. Measurement of the current is then taken over the vertical line, catching only induced current in the horizontal direction -

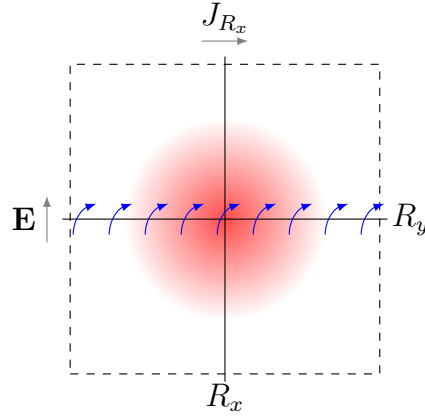


Figure 5.3: A schematic for local Hall measurement. The bottom half of the system has its potential raised with respect to the top half, generating an electric field acting over the horizontal line. Current is measured between the left and right halves, across the vertical line. The flow of charge is represented in blue, showing the curved path due to the presence of a magnetic field. The red shading shows the region around the crosshair where a nonzero contribution to the marker is measured.

effectively only the Hall current. We expect all nonzero contributions to come from the region around the point where the lines cross - motivating the name ‘crosshair marker’.

As we shall see in the following examples, this argument fails when the system becomes conducting and the gap closes. In a conducting system, current is no longer locally constrained - an electron can easily tunnel across the entire system. Thus we expect that the marker will be nonzero even very far from the crosshair. Consequently, as well as the central peak, we expect to see an additional term wherever conducting edge states are present, at any distance from the crosshair.

5.2.1 Exact derivation

We measure the local contribution to current across the line $x = R_x$ in the adiabatic regime. The current will be expressed in the form of a conductivity $J = \sigma E$, where σ is to be determined. Our starting point is the expectation value of the current operator defined in §5.1.3 for the case of a filled band of states,

$$\langle J_{R_x}^A(\mathbf{r}) \rangle = \frac{1}{2} \text{Tr} [P \{ \delta_{\mathbf{r}}, J_{R_x}^A \}], \quad (5.2.1)$$

where P projects onto a fully occupied band of states. By cycling terms in the trace and using the identity $\mathbb{1} = P + Q$, we can split this expression into two terms

$$\begin{aligned} \langle J_{R_x}^A(\mathbf{r}) \rangle &= \text{Tr}_{\mathbf{r}} (P J_{R_x}^A P) \\ &+ \frac{1}{2} \text{Tr}_{\mathbf{r}} (P J_{R_x}^A Q + Q J_{R_x}^A P), \end{aligned} \quad (5.2.2)$$

where $\text{Tr}_{\mathbf{r}} := \text{Tr}(\delta_{\mathbf{r}} \dots)$ denotes a local trace over the internal degrees of freedom at site $\mathbf{r}$. We treat these two terms separately. The first term is a local marker of Chern number whereas

the second term vanishes. Considering only the second term, let us insert the exact form of $J_{R_x}^A = -[[\dot{P}, P], \vartheta_{R_x}]$ and use the Jacobi identity,

$$\begin{aligned} \text{Tr}_{\mathbf{r}} (P J_{R_x}^A Q) + h.c. &= \text{Tr}_{\mathbf{r}} \left(P \left[[P, \vartheta_{R_x}], \dot{P} \right] Q \right) \\ &+ \text{Tr}_{\mathbf{r}} \left(P \left[[\vartheta_{R_x}, \dot{P}], P \right] Q \right) + h.c. \end{aligned} \quad (5.2.3)$$

Using the identities $P[P, \vartheta_{R_x}] = [P, \vartheta_{R_x}]Q$ and $P\dot{P} = \dot{P}Q$, we see that the first term here vanishes. We are left with the second term which can be written as

$$\text{Tr}_{\mathbf{r}} (P J_{R_x} Q) + h.c. = -\text{Tr}_{\mathbf{r}} \left(P [\vartheta_{R_x}, \dot{P}] Q \right) + h.c. \quad (5.2.4)$$

Using $Q = \mathbb{1} - P$, we may re-express this in the form,

$$\text{Tr}_{\mathbf{r}} (P J_{R_x} Q) + h.c. = -\text{Tr}_{\mathbf{r}} \left(P [\vartheta_{R_x}, \dot{P}] \right) + h.c. \quad (5.2.5)$$

Now, let us insert the exact form for $\dot{P}$ for a system undergoing adiabatic application of the step-function electric field defined in §5.1.3: $\dot{P} = i\dot{A}[\vartheta_{R_y}, P]$. We get a final expression

$$\begin{aligned} \text{Tr}_{\mathbf{r}} (P J_{R_x} Q) + h.c. \\ = -i\dot{A} \text{Tr}_{\mathbf{r}} \left(P [\vartheta_{R_x}, [\vartheta_{R_y}, P]] \right) + h.c. \end{aligned} \quad (5.2.6)$$

Writing out the terms in the trace explicitly, we get

$$\begin{aligned} i\text{Tr}_{\mathbf{r}} \left(P [\vartheta_{R_x}, [\vartheta_{R_y}, P]] \right) &= i\text{Tr}_{\mathbf{r}} (P \vartheta_{R_x} \vartheta_{R_y} P) \\ &- i\text{Tr}_{\mathbf{r}} (P \vartheta_{R_x} P \vartheta_{R_y}) \\ &- i\text{Tr}_{\mathbf{r}} (\vartheta_{R_x} P \vartheta_{R_y} P) \\ &+ i\text{Tr}_{\mathbf{r}} (P \vartheta_{R_x} \vartheta_{R_y}). \end{aligned} \quad (5.2.7)$$

The step functions commute with one another, and with $\delta_{\mathbf{r}}$. This means that every constituent term in eqn. 5.2.6 is anti-Hermitian due to the factor of i . Thus the whole expression vanishes when the Hermitian conjugate is added,

$$\text{Tr}_{\mathbf{r}} (P J_{R_x} Q) + h.c. = 0. \quad (5.2.8)$$

Now, let us return to eqn 5.2.2. We have shown that the second term vanishes, so we are left with

$$\langle J_{R_x}(\mathbf{r}) \rangle = \text{Tr}_{\mathbf{r}} (P J_{R_x} P) \quad (5.2.9)$$

$$= i\text{Tr}_{\mathbf{r}} (P [K, \vartheta_{R_x}] P). \quad (5.2.10)$$

We use expression 5.1.14 to insert the exact form of K

$$\langle J_{R_x}(\mathbf{r}) \rangle = i\dot{A} \text{Tr}_{\mathbf{r}} (P \vartheta_{R_x} Q \vartheta_{R_y} P) + h.c. \quad (5.2.11)$$

Using the fact that $\dot{A} = -E$, we see that this expectation value has taken the form of a conductance, $\langle J \rangle = \sigma E$. We also use $Q = \mathbb{1} - P$ to get the conductance in the form

$$\sigma(\mathbf{r}; \mathbf{R}) = 2\Im \text{Tr}_{\mathbf{r}} (P \vartheta_{R_x} P \vartheta_{R_y} P). \quad (5.2.12)$$

Finally we insert a factor of 2π to get the final expression for the marker

$$C(\mathbf{r}; \mathbf{R}) = 4\pi \Im \text{Tr}_{\mathbf{r}} (P \vartheta_{R_x} P \vartheta_{R_y} P). \quad (5.2.13)$$

This quantity is analogous to the Chern marker presented in appendix C of [8], although where Kitaev used a global trace, here we have a local trace. Kitaev's marker was shown to exactly equal the Chern number in an infinite system. For finite system size, however, his marker vanishes due to edge-state contributions. By replacing the global trace with a local trace, we are here able to separate the bulk contribution from the edge contribution, allowing for the marker to be calculated in a finite system. We can extract the quantised Chern number by only summing over the bulk contributions – found at $\mathbf{r}$ close to $\mathbf{R}$ – and ignoring the edge-state terms at $\mathbf{r}$ far from $\mathbf{R}$.

We note two important caveats. Firstly the marker is only strictly well-defined as the sum of the bulk contributions around $\mathbf{R}$, rather than as a function of $\mathbf{r}$. This is because there is a degree of arbitrariness to how the current was made local to a point in definition 5.1.10, since the purpose is to separate the bulk and edge contributions before summing only the bulk terms. Physically, the summed marker corresponds to the total current measured in the bulk over the line at $x = R_x$.

The second caveat is that the adiabatic current defined in eqn. 5.1.16 only captures the current generated by the change in P over time. However, it is also possible for a system to have persistent currents, i.e. flow between occupied states that does not change the projector itself. Since this type of circulating current cannot change the number density at any site, it does not contribute to the net flow of electrons into a given region. Despite this, it can still contribute terms to the current defined in eqn. 5.1.10. Thus we must also require that the system has no persistent currents at any point in the adiabatic time evolution. This effect is explained in detail in appendix ??.

Finally it is worth mentioning that the crosshair marker can be used to calculate the Chern marker, $\mathfrak{C}$, given in [11] according to

$$\mathfrak{C}(\mathbf{r}) = \int d^2\mathbf{R} C(\mathbf{r}; \mathbf{R}), \quad (5.2.14)$$

where the integral is over the whole system. The proof of this is given in appendix ??.

5.3 Examples

Examples are calculated for two sets of topological quantum systems. The first will be the Qi-Wu-Zhang (QWZ) model on a square lattice, the simplest example of a Chern insulator with only nearest neighbour interactions [23, 24]. The second example will be an extension of the QWZ model to amorphous lattices.

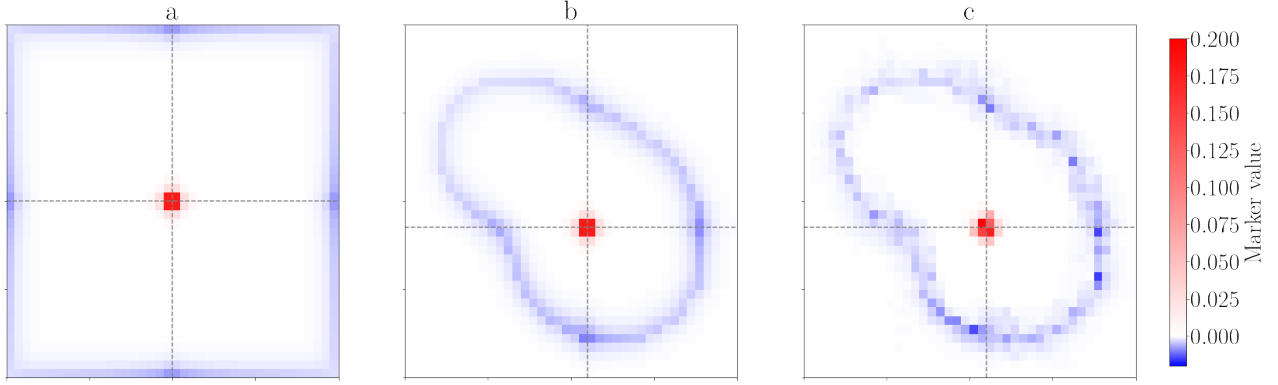


Figure 5.4: Crosshair marker, $C(\mathbf{r}; \mathbf{R})$, plotted as a function of $\mathbf{r}$ for three example systems on 40×40 lattices. Position of the crosshair (determined by $\mathbf{R}$) is indicated with dashed lines, where $\mathbf{R}$ indicates the point where the lines meet. a) A uniform QWZ system with $u = 1.6$. b) A compound system with $u = 1.6$ inside the central region and $u = 2.6$ outside. The boundary of the region follows the blue ring, which indicates the presence of conducting edge states. c) A compound system, as in (b), however u has an added per-site Gaussian noise term with standard deviation $\sigma = 0.5$.

5.3.1 Qi-Wu-Zhang Model

We work on a square lattice of size (L_x, L_y) , where site positions are given by $\mathbf{r}_i = (n_x, n_y) \in \mathbb{Z}^2$. Each site has two internal degrees of freedom. The Hamiltonian is

$$\begin{aligned}
 H = & \sum_{\mathbf{r}} |\mathbf{r} + \mathbf{1}_x\rangle \langle \mathbf{r}| \otimes \frac{\sigma_z + i\sigma_x}{2} + h.c. \\
 & + \sum_{\mathbf{r}} |\mathbf{r} + \mathbf{1}_y\rangle \langle \mathbf{r}| \otimes \frac{\sigma_z + i\sigma_y}{2} + h.c. \\
 & + \sum_{\mathbf{r}} |\mathbf{r}\rangle \langle \mathbf{r}| \otimes u \sigma_z,
 \end{aligned} \tag{5.3.1}$$

where σ_i are the Pauli matrices. The parameter u determines the topological properties of the Hamiltonian. In a uniform system, the Chern number is -1 for $-2 < u < 0$, $+1$ for $0 < u < 2$ and zero otherwise. In the following examples we break translational symmetry by allowing u to vary on a per-site basis, $u \rightarrow u_{\mathbf{r}}$.

We calculate the crosshair marker for three example systems, shown in figure 5.4. The first system is a uniform lattice with open boundaries. In the second we show a compound system with open boundaries and a central region that we expect to have Chern number $+1$, surrounded by a region with Chern number 0 . In all cases, the crosshair is placed in the bulk of the topological region and the expected Chern number is $+1$. Close to the crosshair, there is a positive contribution from the local conductance as derived in §5.2. Around the boundary of the topological region there is a negative contribution due to the presence of edge states. The overall trace of the marker vanishes, so this negative contribution sums to the same magnitude as the central peak.

To demonstrate the quantisation of the marker, we sum the value enclosed within a radius around the central peak. When the radius is sufficiently large that it captures the whole peak the sum equals to the Chern number. If the radius includes contributions from the conducting

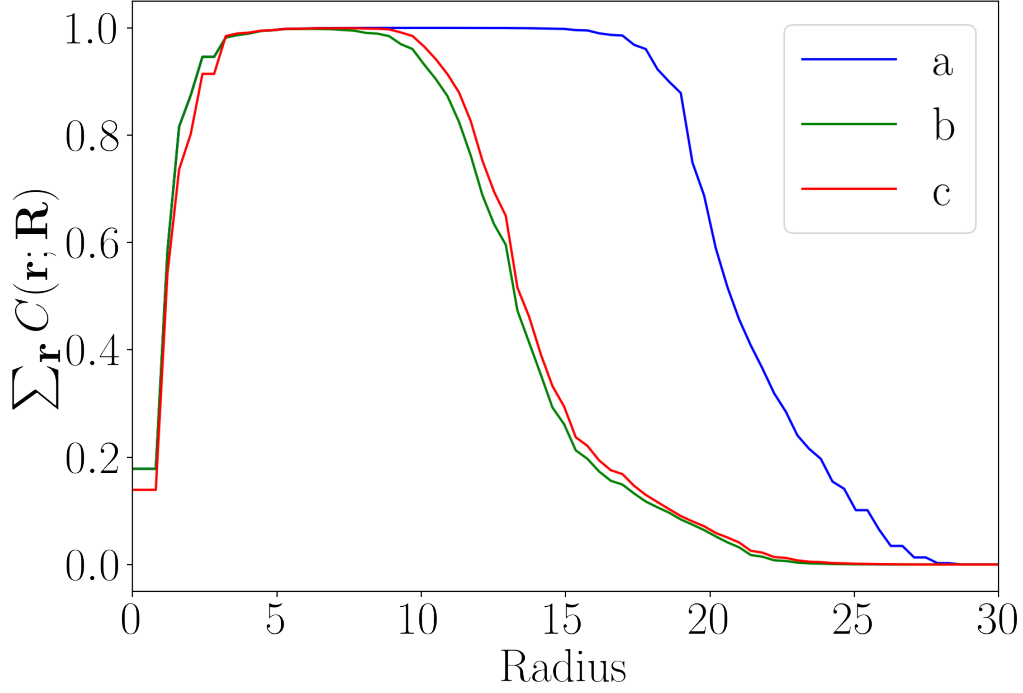


Figure 5.5: Plots of the sum of the crosshair marker in $\mathbf{r}$ for a circle of a given radius around the crosshair position $\mathbf{p}$. Plots are shown for the three examples in fig. 5.4. The maximum values reached in each case are 0.99997, 0.9981 and 0.9989 respectively.

edge states, the value will be reduced. This is demonstrated in fig. 5.5, where we have plotted the sum against the radius of the region being summed over. To estimate the quantised sum of the marker, we take the maximum value in this range, and see that this estimate is quantised to around one part in 10^3 .

5.3.2 Amorphous Qi-Wu-Zhang Model

We now look at extending the QWZ model beyond a grid to arbitrary amorphous lattices, providing an example of a system that is completely inaccessible to traditional methods for calculating Chern number. We prepare a set of arbitrary lattices using a Voronoi construction, identical to the construction of a Wigner-Seitz cell [25, 26]. Starting with a set of random points, we may generate a cell for each point containing the region of space closer to that point than any other. A lattice is created by taking the barrier between Voronoi cells as the edges, and the vertices as the points where three adjacent Voronoi cells meet. This generates a lattice where every point is trivalent. An example lattice is shown in figure 5.6.

Once a lattice has been chosen, the Hamiltonian is constructed using a smooth continuation of the QWZ model to account for edges that can point in an arbitrary direction. The Hamiltonian is given by

$$\begin{aligned}
 H = & \sum_{i,j} |\mathbf{r}_i\rangle \langle \mathbf{r}_j| \otimes h_{\theta_{i,j}} + h.c. \\
 & + \sum_i |\mathbf{r}_i\rangle \langle \mathbf{r}_i| \otimes u \sigma_z.
 \end{aligned} \tag{5.3.2}$$

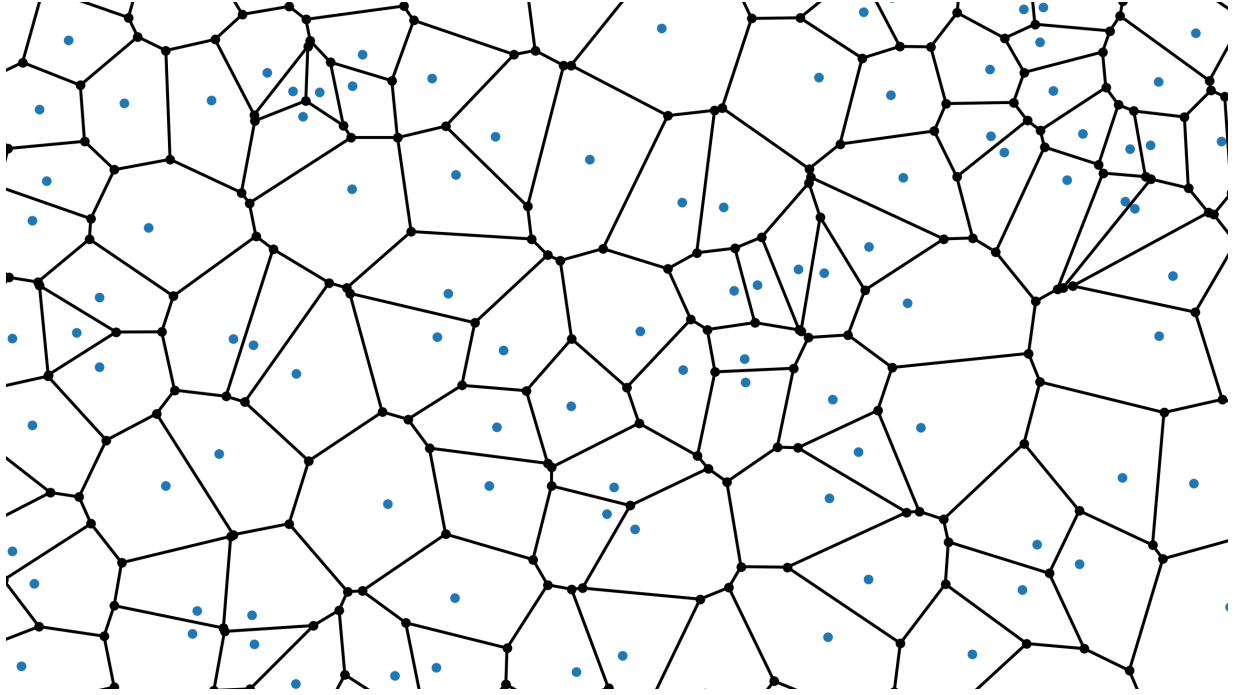


Figure 5.6: An example section of amorphous lattice. Shown in blue are a set of points that have been randomly generated. In black we see the resulting Voronoi lattice, where each cell encloses the region of space that is closer to a given blue point than any other. Our sites (shown in black) are placed at the corners where three cells meet and connected with an edge along the boundary between two cells.

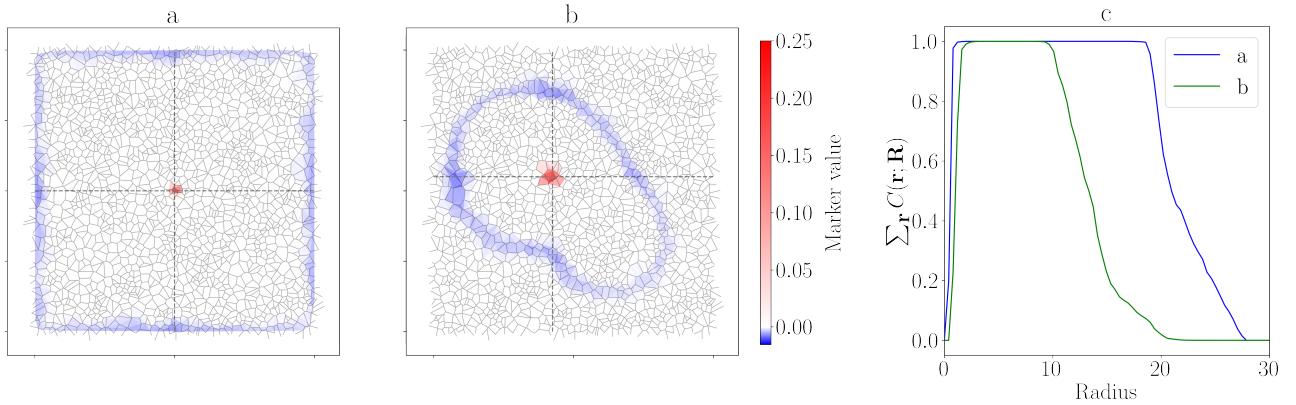


Figure 5.7: Crosshair marker, $C(\mathbf{r}; \mathbf{R})$, plotted as a function of $\mathbf{r}$ for two example amorphous systems. As in fig. 5.4, the position of the crosshairs (determined by $\mathbf{R}$) is indicated with dashed lines. Both cases are for the same lattice with 3200 sites, decorated with a different Hamiltonian. a) The crosshair marker for the system with uniform $u = -1$. b) Crosshair marker for a compound amorphous system with $u = -1$ inside the central region and $u = -2$ outside. c) Plots of the sum of the crosshair markers for the two given systems. Maximum values reached in are 1.0007, and 1.00002 respectively

Here, rather than the standard x and y -hopping matrices from eqn. 5.3.1, we use a matrix h_θ that depends on the angle θ that the edge $\mathbf{r}_j \rightarrow \mathbf{r}_i$ makes with the x -axis. This is defined as

$$h_\theta = \frac{1}{2} \begin{pmatrix} 1 & ie^{-i\theta} \\ ie^{i\theta} & -1 \end{pmatrix}. \quad (5.3.3)$$

Inserting into this expression $\theta = 0$ or $\theta = \frac{\pi}{2}$, we recover the standard QWZ hopping matrices. Also note that $h_{\theta+\pi} = h_\theta^*$.

As before, the parameter u determines whether the system is in a topological or trivial state, although the phase does not have the same dependence on u as before. By considering the band structure as a function of u , we determine numerically that the system is gapped and topological for $-3 < u < -1$. A justification for this is presented in appendix ??.

In fig. 5.7 we show crosshair markers for two example systems, as well as a plot of sums of the markers against the radius of the region summed over. As before, the bulk sum is quantised to around one part in 10^3 .

5.4 Conclusion and Outlook

We have developed a physically motivated local marker for the Chern number of a two-dimensional quantum system. The marker was derived as a local Hall conductivity for current around a crosshair in the bulk. We present a rigorous expression for this conductance for non-interacting electrons on the lattice, and show that it is equivalent to the real-space Chern number presented in [8], although we make a slight modification to ensure it is nonzero for finite-size systems. Mathematically, the marker plays a similar role to existing methods such as Chern marker [11] and Bott index [12] – locally indicating the topological phase of the system – however unlike these markers, the crosshair marker is precisely quantised in the bulk.

The arguments used to derive the crosshair marker are essentially a prescription for measuring the quantity in the lab: a step-function potential is used to excite current across the system in the vertical direction. The cross-conductance is then measured in the horizontal direction. An obvious next step will be to look at experimentally verifying the proposed method.

Until recently, local markers were developed in a non-interacting context as a property derived from the projector onto a band of states. In interacting systems, where the ground state does not correspond to a projector onto a set of single-particle states, most existing markers are inapplicable, however early progress has been made in understanding how local topology can be detected in interacting systems [27]. Clearly, in such a system the mathematical formalism presented here does not work, however our physical argument does not depend on the microscopic details of the system. Thus, it would be interesting to re-express the marker in a many-body framework. Similarly, we expect that the physical definition of the marker could provide a clue to understanding the role that local topology has to play in systems out of equilibrium [13, 28, 29].

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Chapter 6

Kitaev's Honeycomb Model

Chapter 7

The Amorphous Kitaev Model

Chapter 8

The Star Lattice Kitaev-Heisenberg Model

Chapter 9

Conclusion and Outlook